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Creasy et al.

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(54) **SNAP-LOCK PHOTOVOLTAIC MODULE MOUNTING SYSTEM**

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(22) Filed: **May 1, 2024**

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Related U.S. Application Data

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(60) Provisional application No. 63/228,261, filed on Aug. 2, 2021, provisional application No. 63/153,326, filed on Feb. 24, 2021.

(51) **Int. Cl.**

H02S 30/10	(2014.01)
F16B 2/06	(2006.01)
F16M 13/02	(2006.01)

(52) **U.S. Cl.**

CPC ***H02S 30/10*** (2014.12); ***F16B 2/06*** (2013.01); ***F16M 13/02*** (2013.01)

(58) **Field of Classification Search**

CPC H02S 30/10; F16B 2/06; F16M 13/02

USPC ...... 24/279, 135 K; 248/218.4–219.4, 218.1, 248/302, 70, 74.1–74.3

See application file for complete search history.

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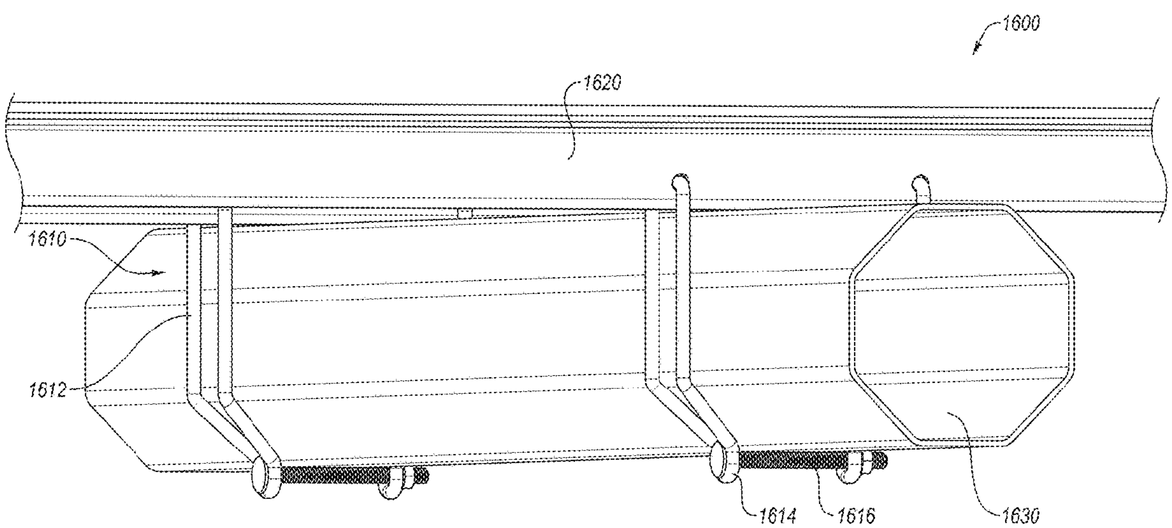
(74) *Attorney, Agent, or Firm* — MASCHOFF BRENNAN

(57) **ABSTRACT**

A photovoltaic (PV) module clamp may be configured to interface with a torque tube and a PV module rail fixedly coupled to a PV module. The PV module clamp may include a seating portion made of two or more lateral walls and a base surface. A clamp body of the PV module clamp may be coupled to the seating portion. The clamp body may include a shape corresponding to a cross-sectional shape of the torque tube. The PV module clamp may include a fastening feature for interlocking the PV module clamp with the PV module rail responsive to the PV module rail being seated in the seating portion of the PV module clamp.

8 Claims, 27 Drawing Sheets



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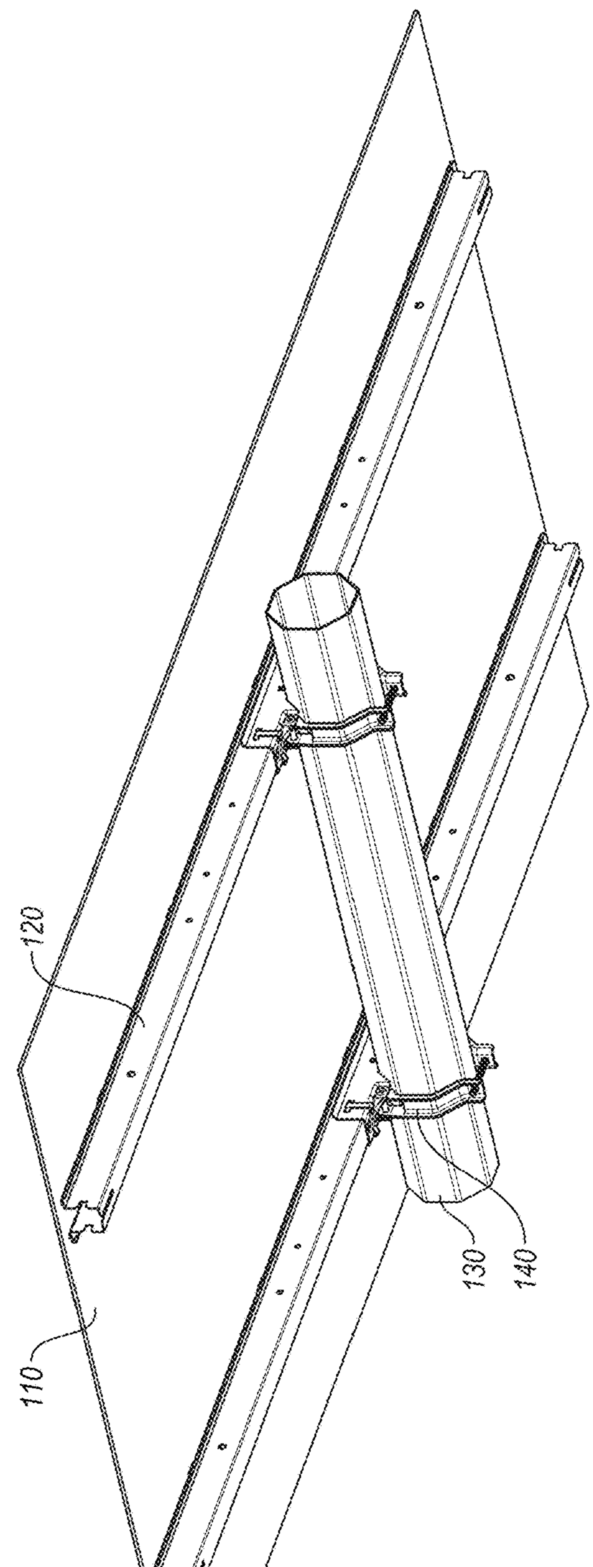

FIG. 1A

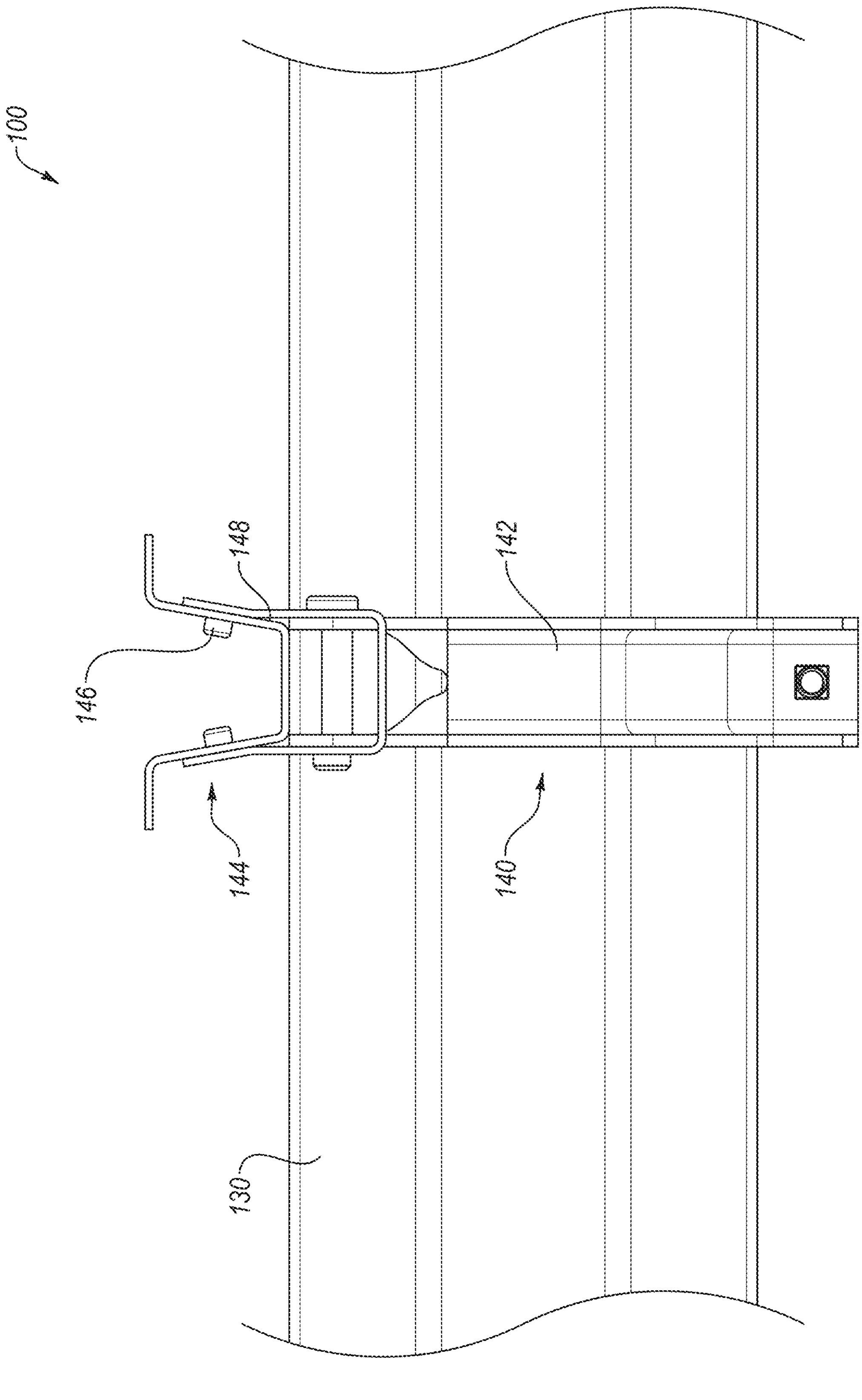

FIG. 1B

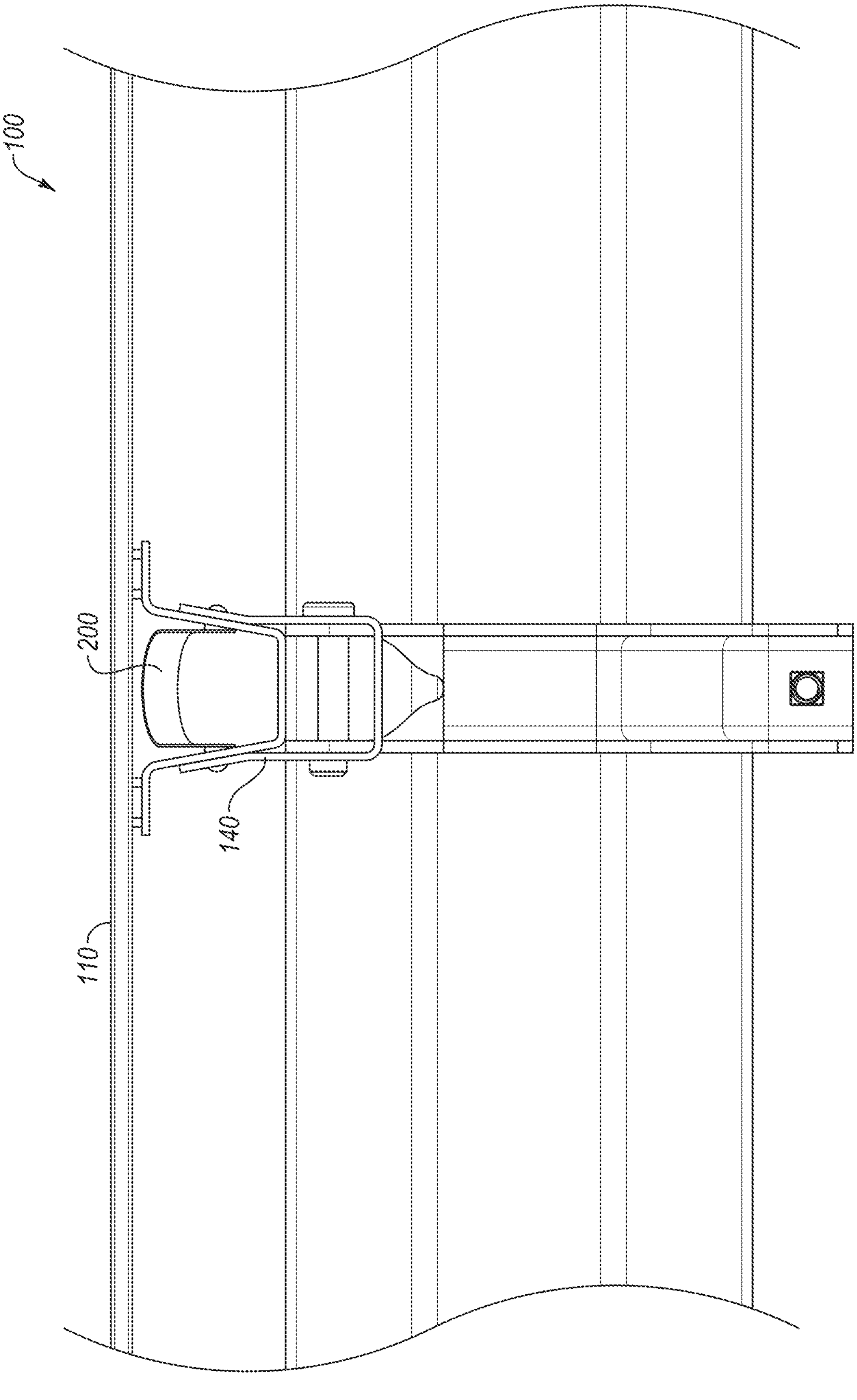

FIG. 1C

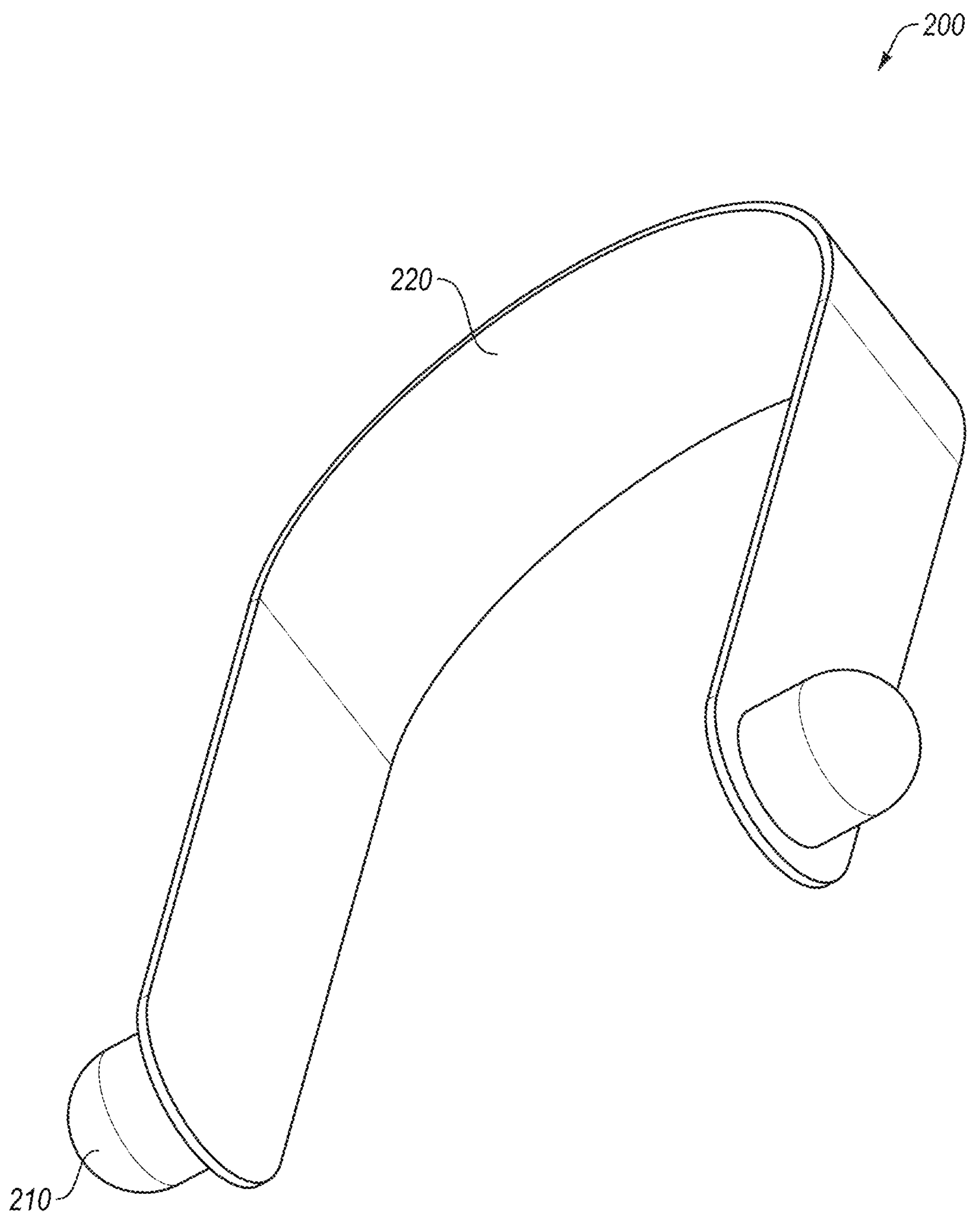

FIG. 2

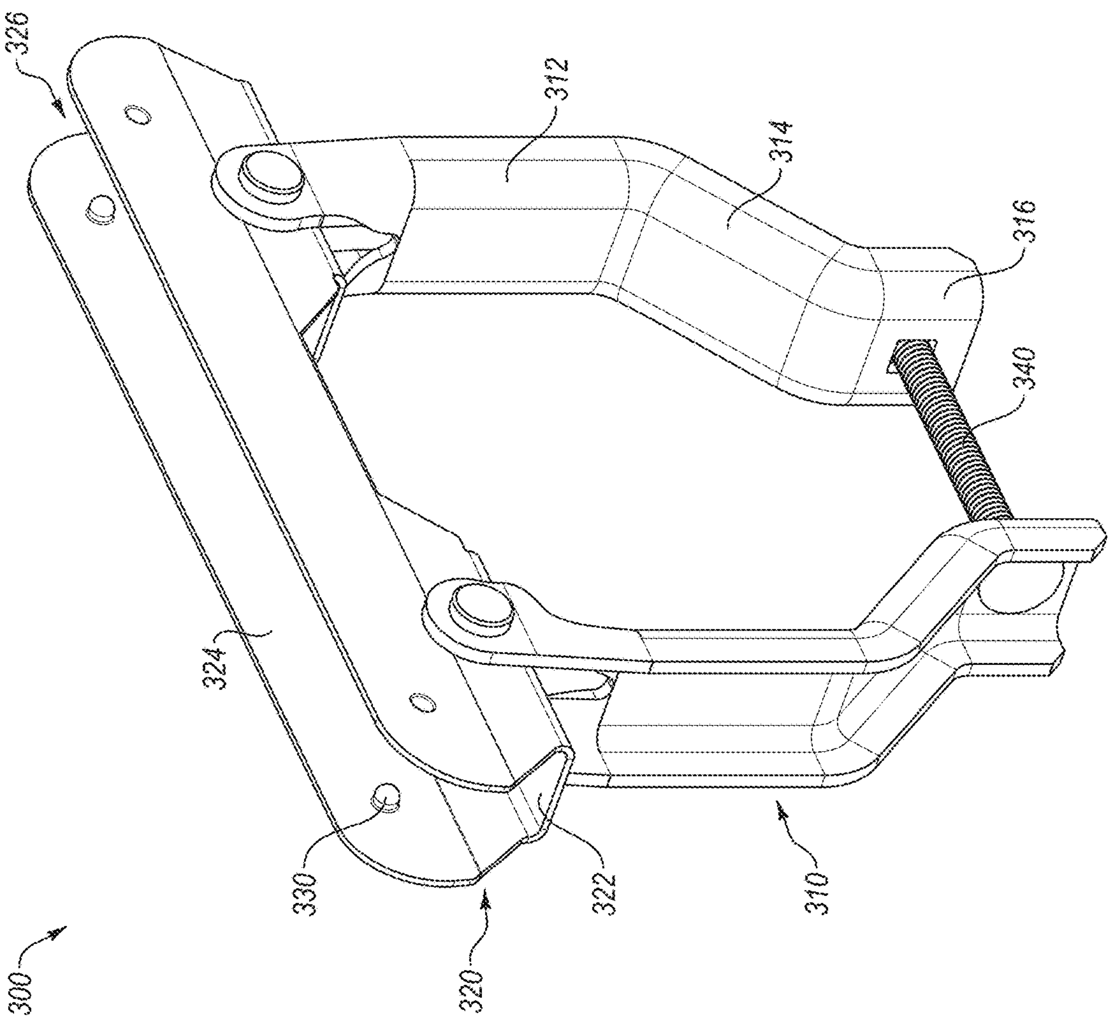

FIG. 3A

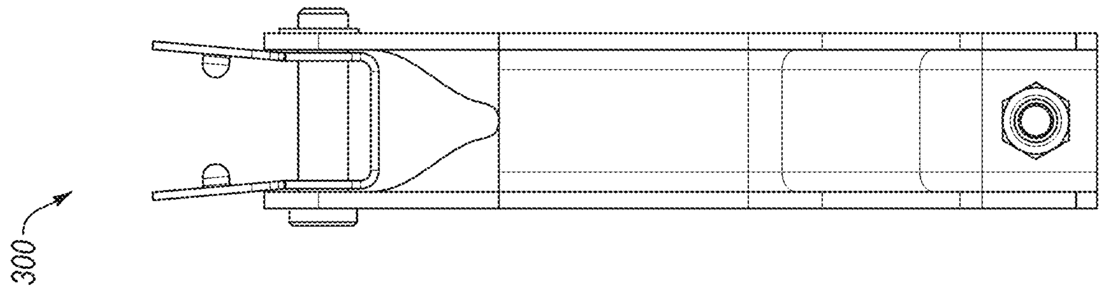

FIG. 3C

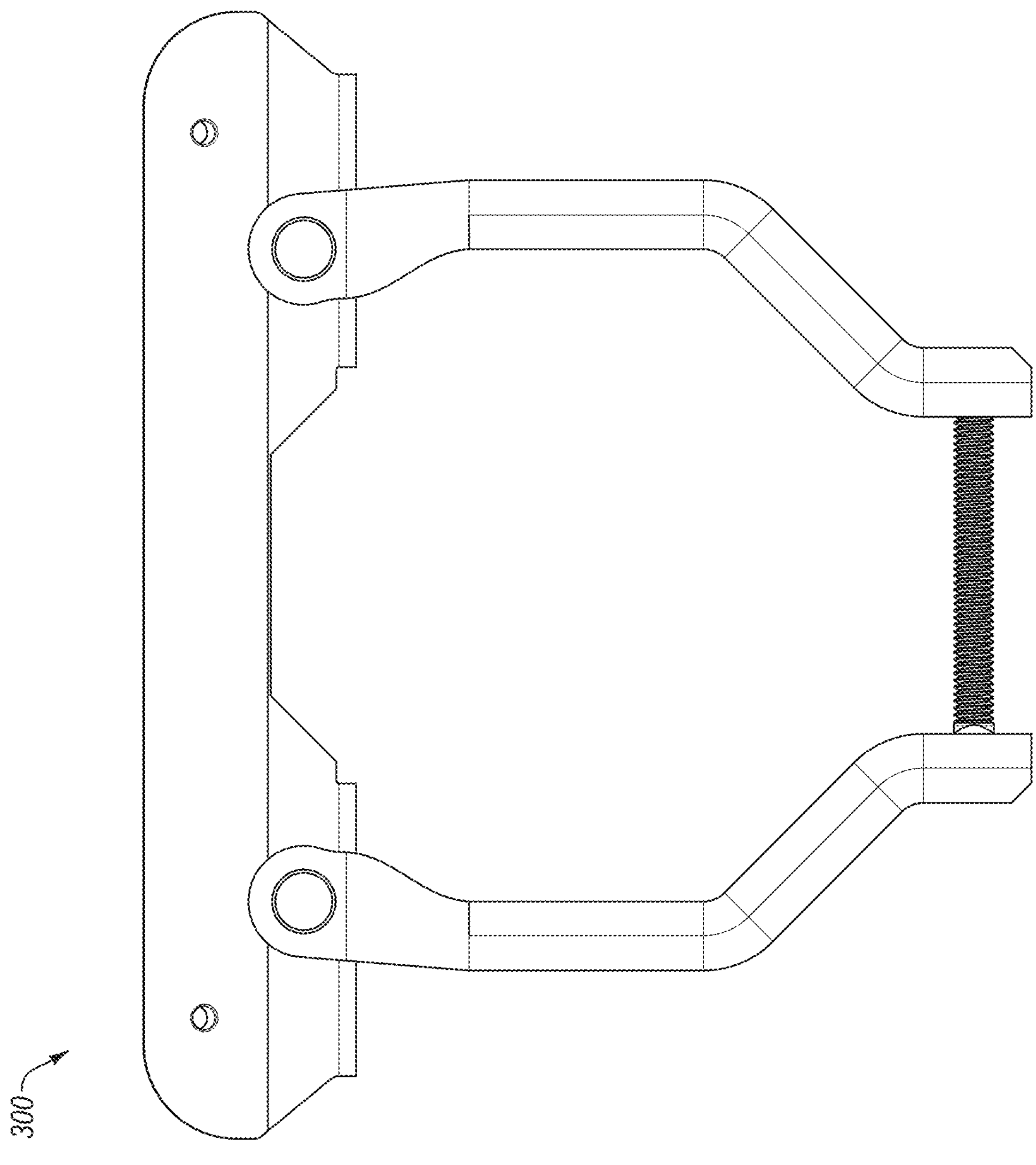

FIG. 3B

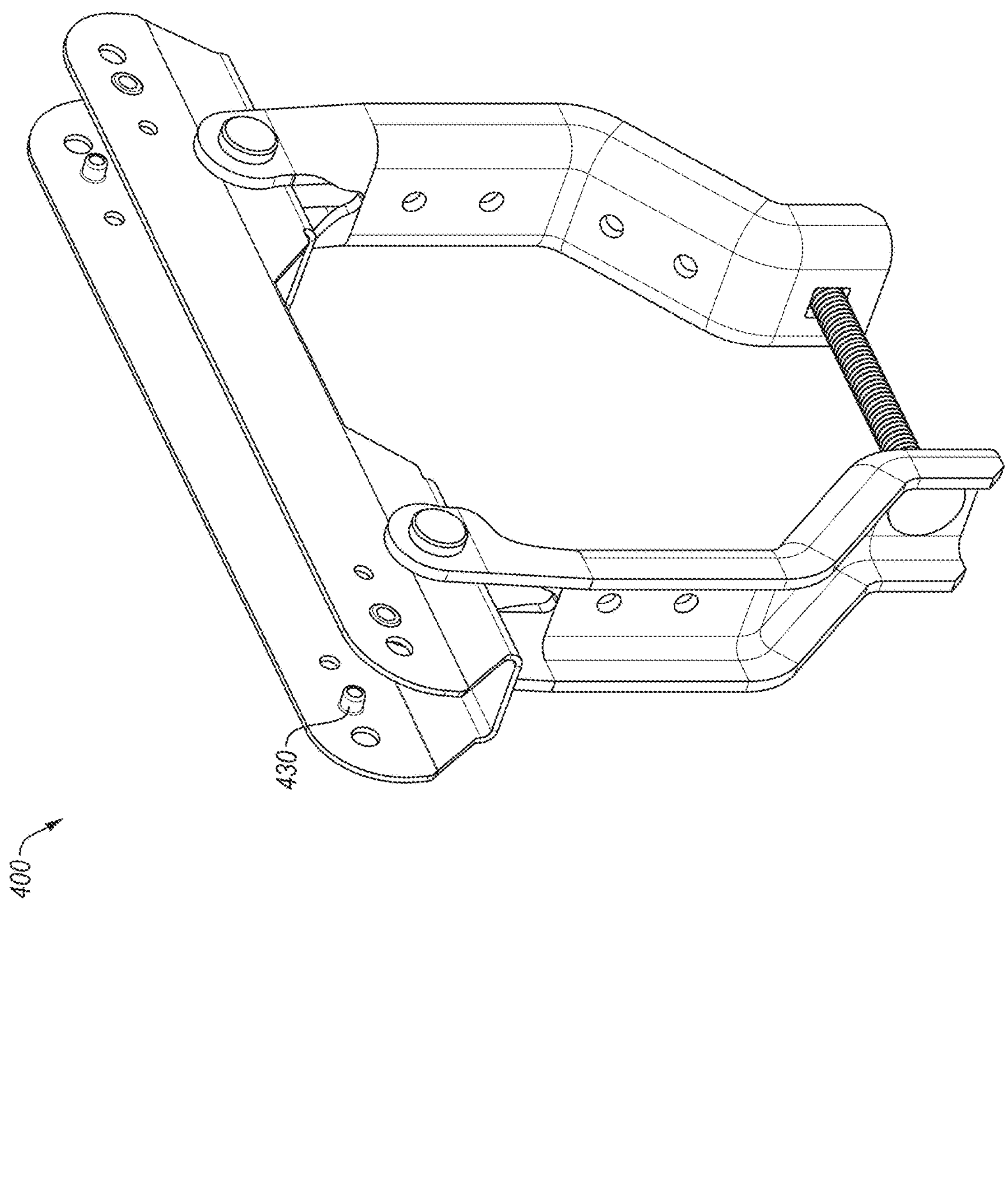

FIG. 4A

400

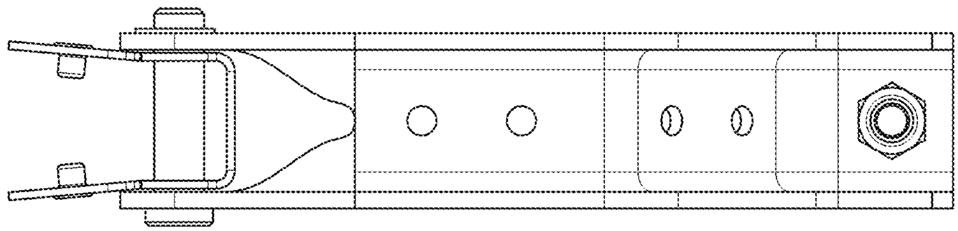

FIG. 4C

400

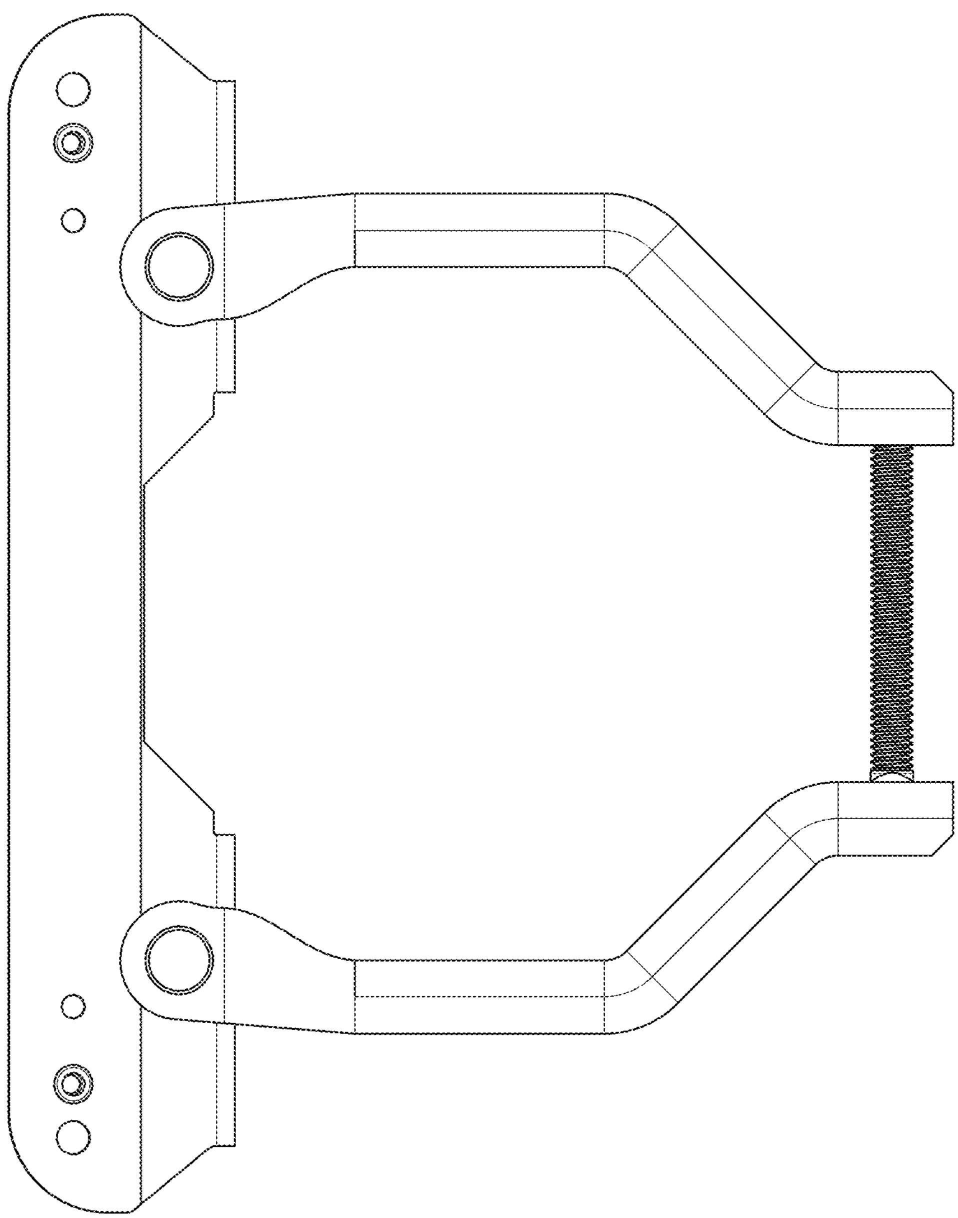

FIG. 4B

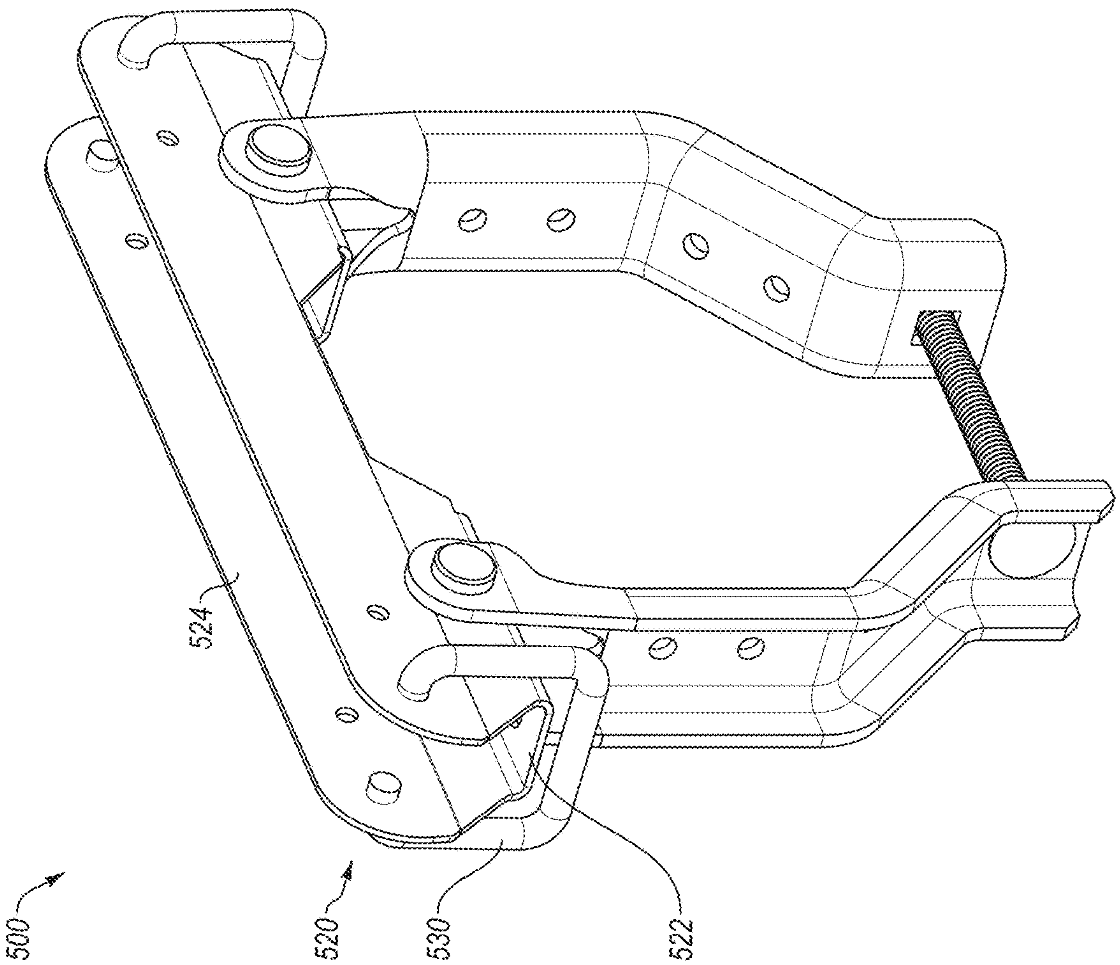

FIG. 5A

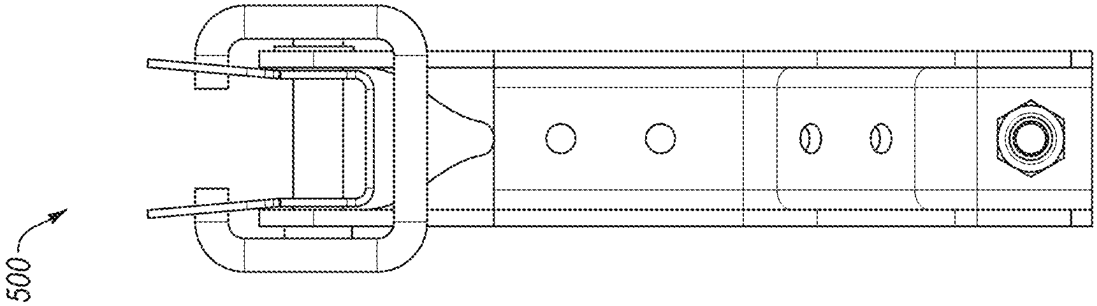

FIG. 5C

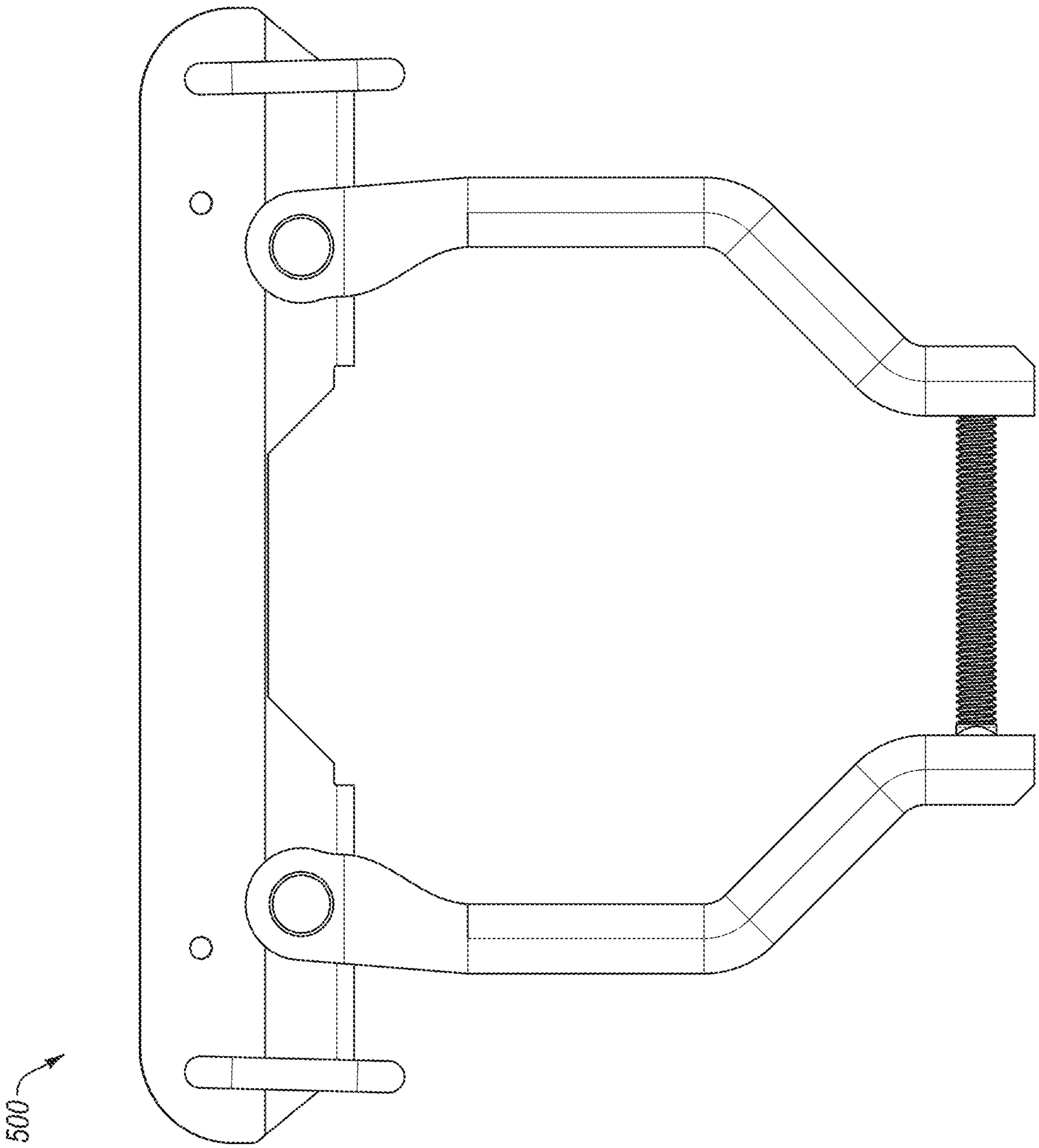

FIG. 5B

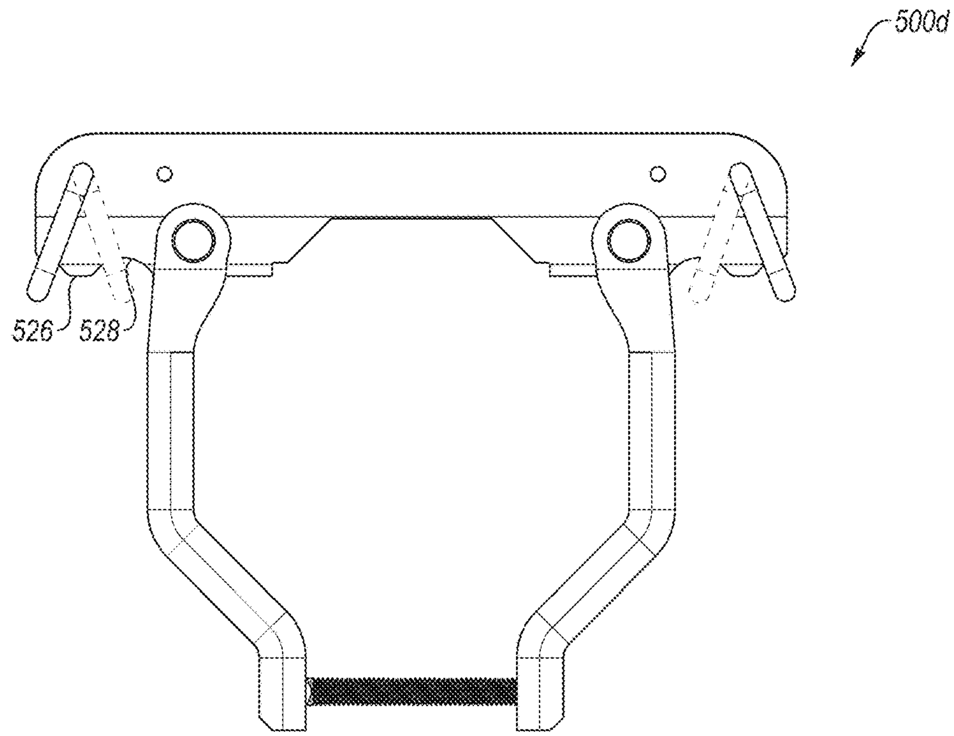

FIG. 5D

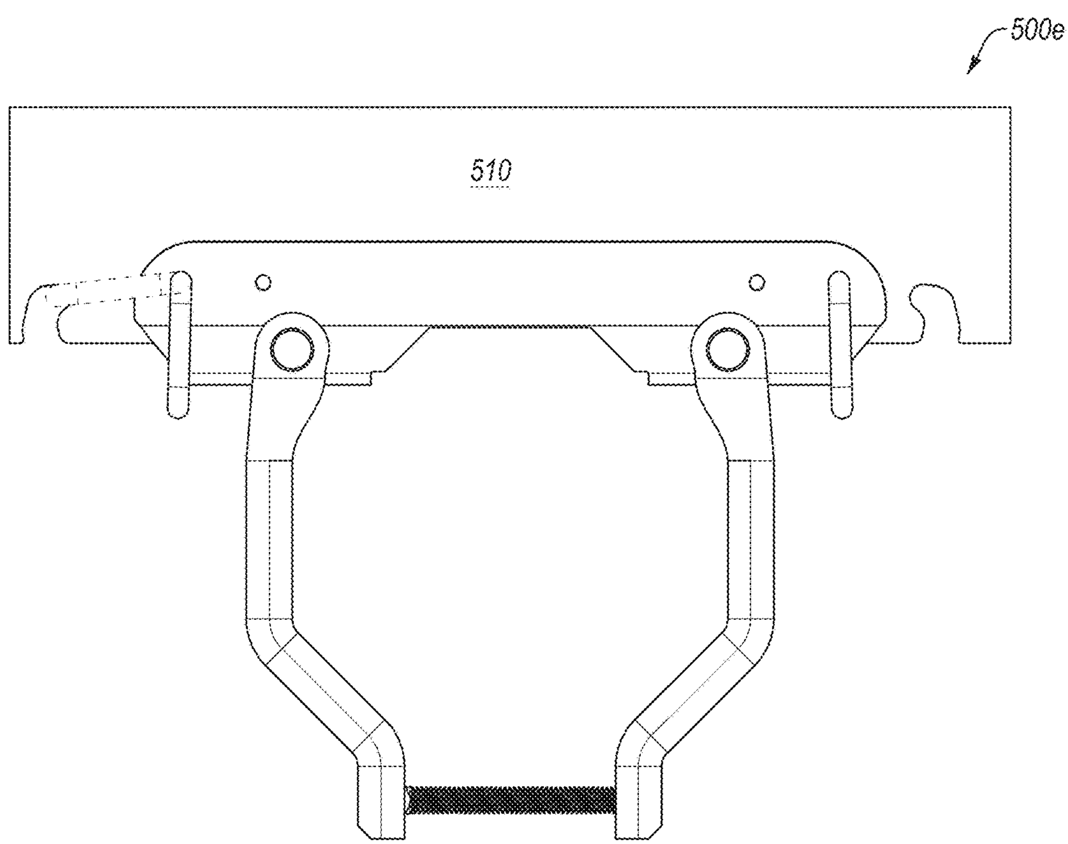

FIG. 5E

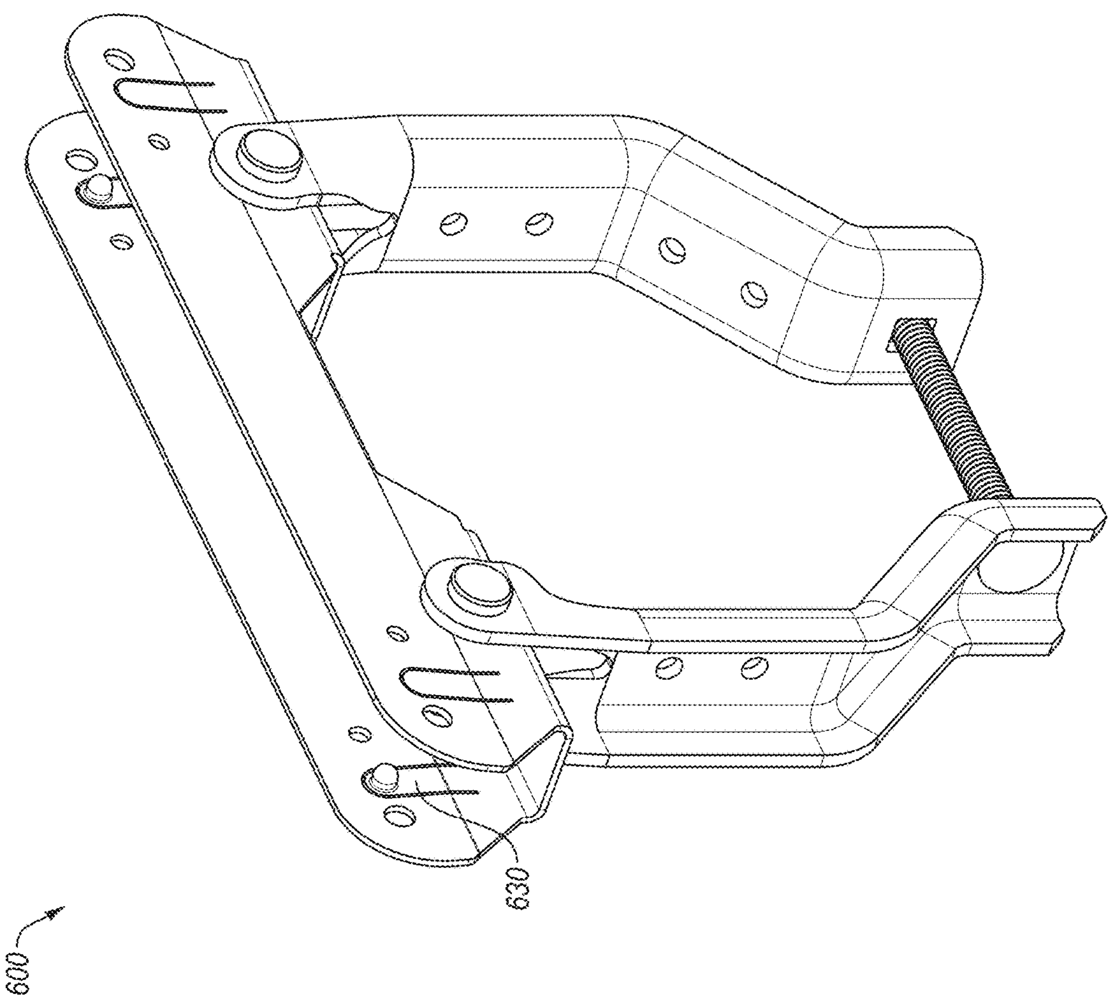

FIG. 6

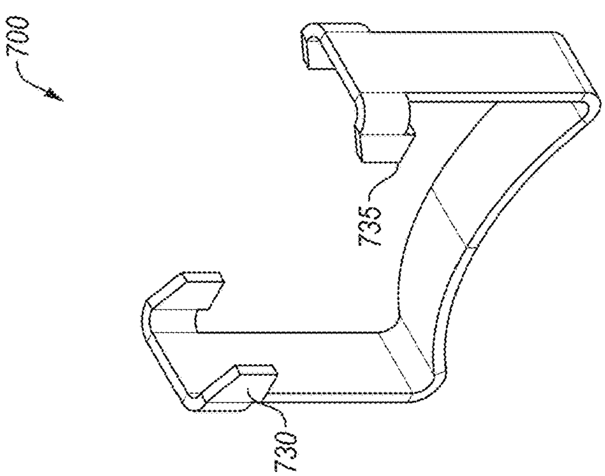

FIG. 7A

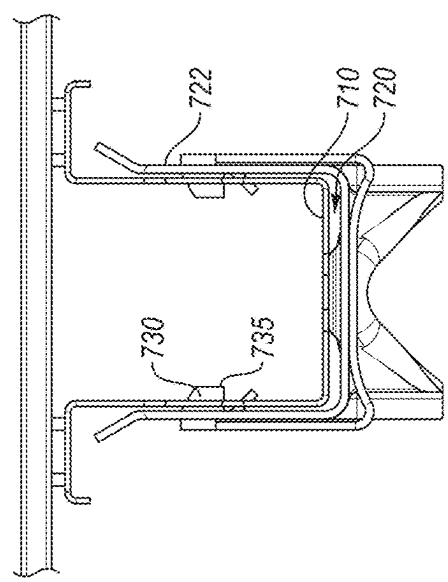

FIG. 7B

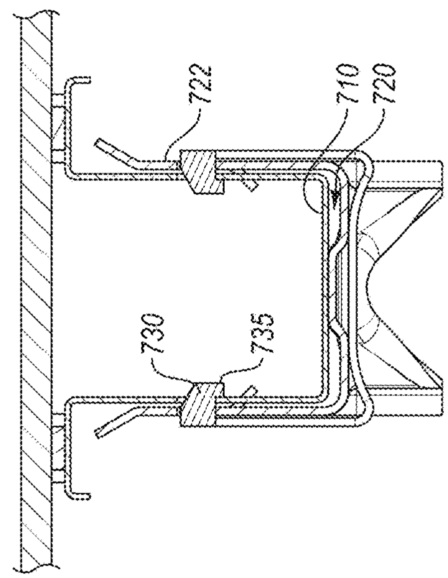

FIG. 7C

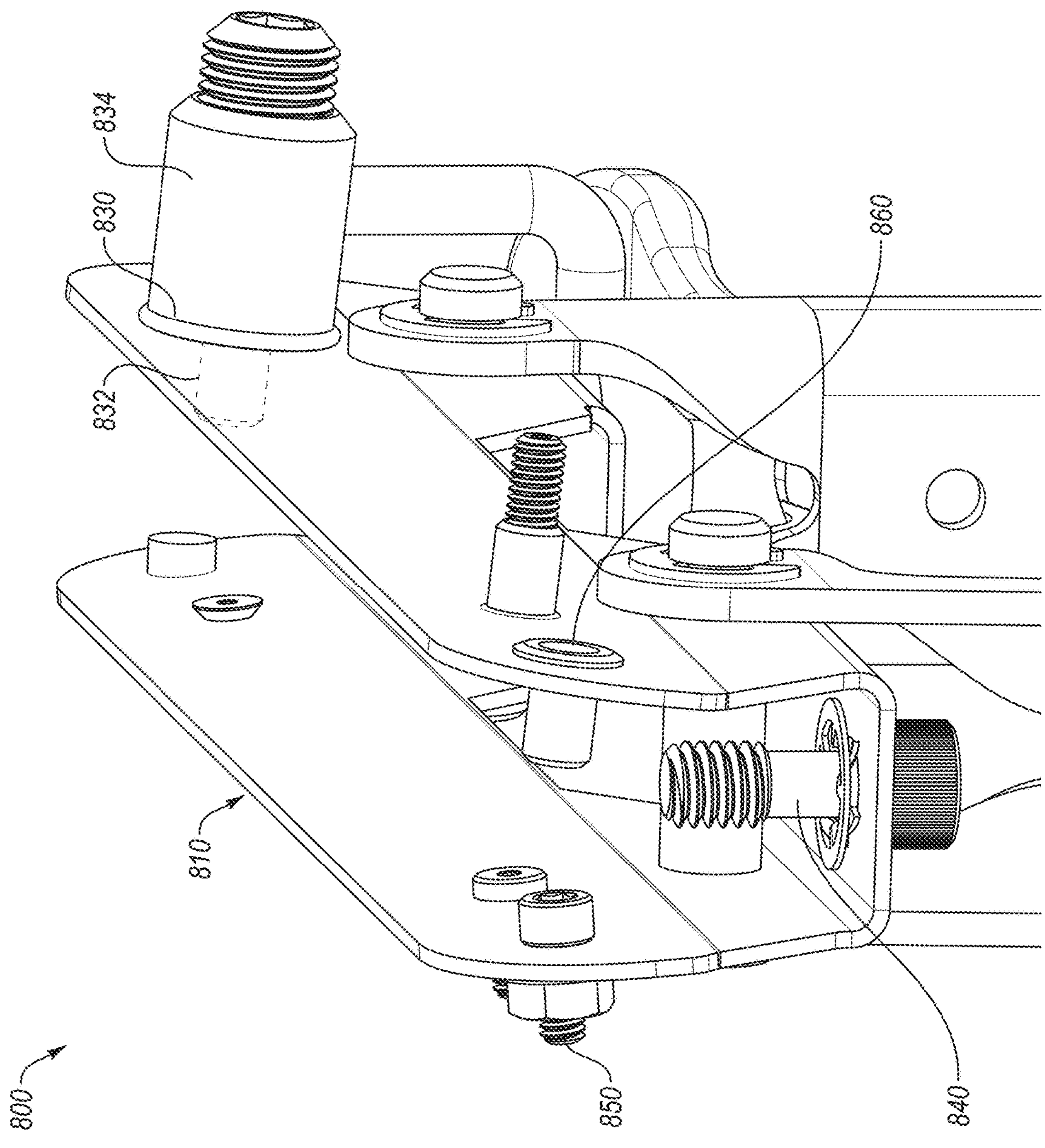

FIG. 8

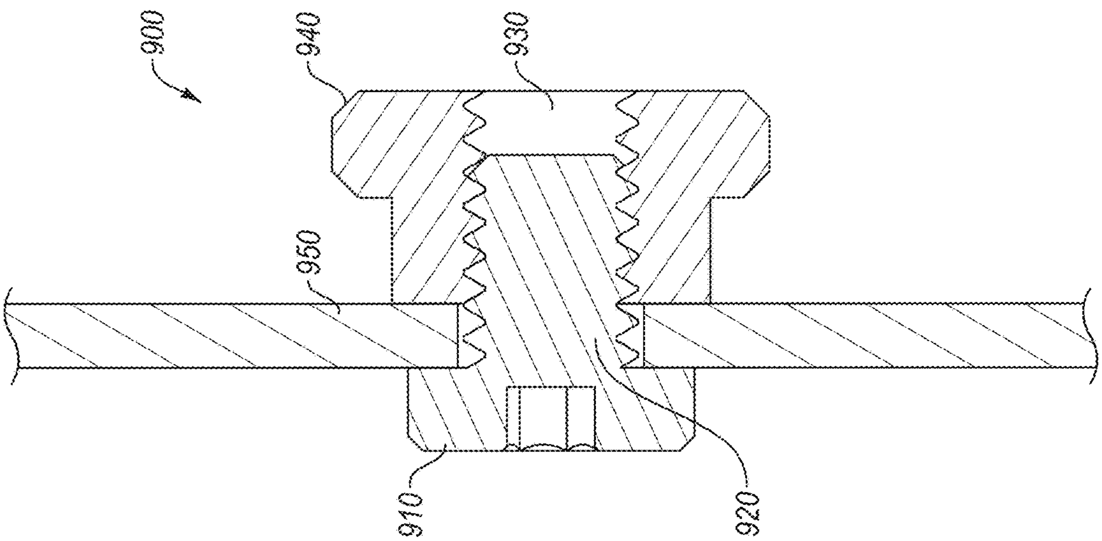

FIG. 9

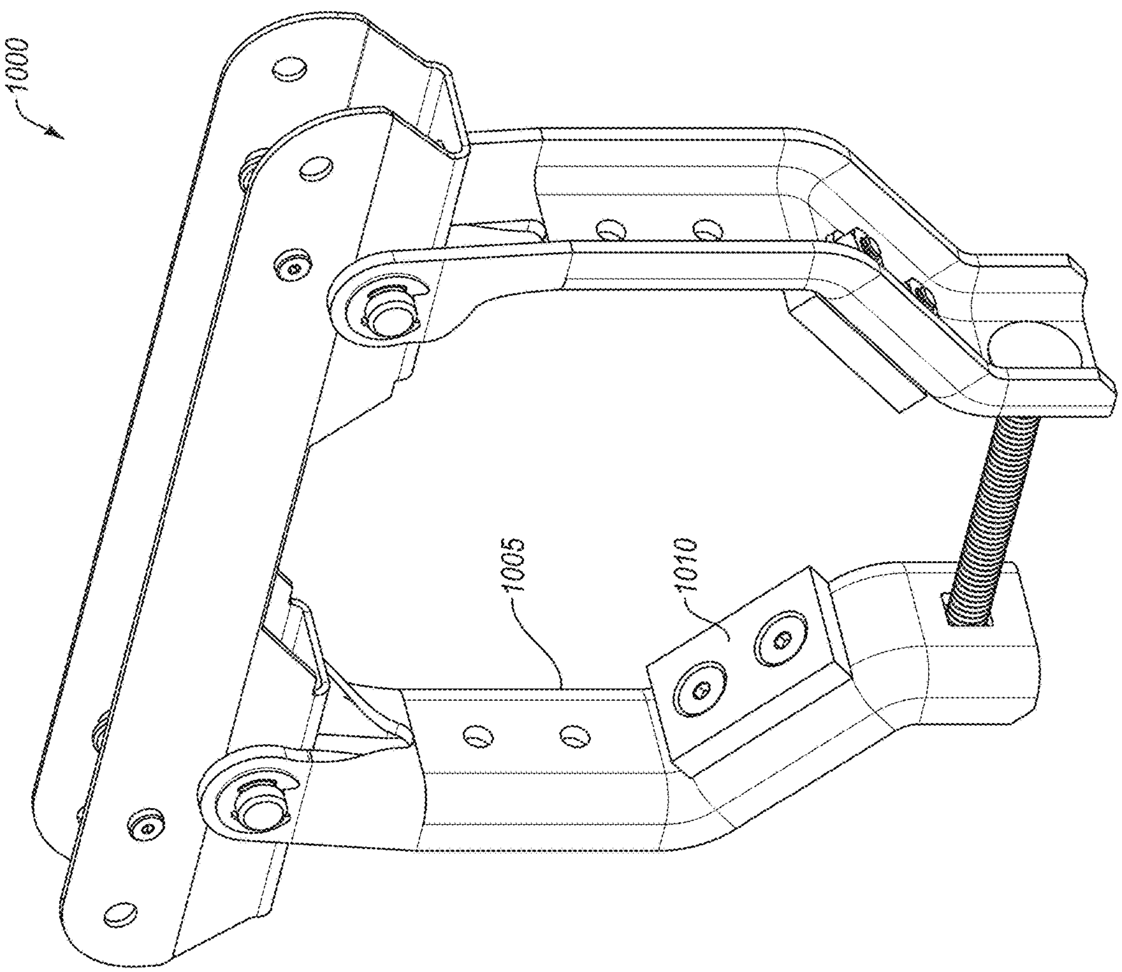

FIG. 10A

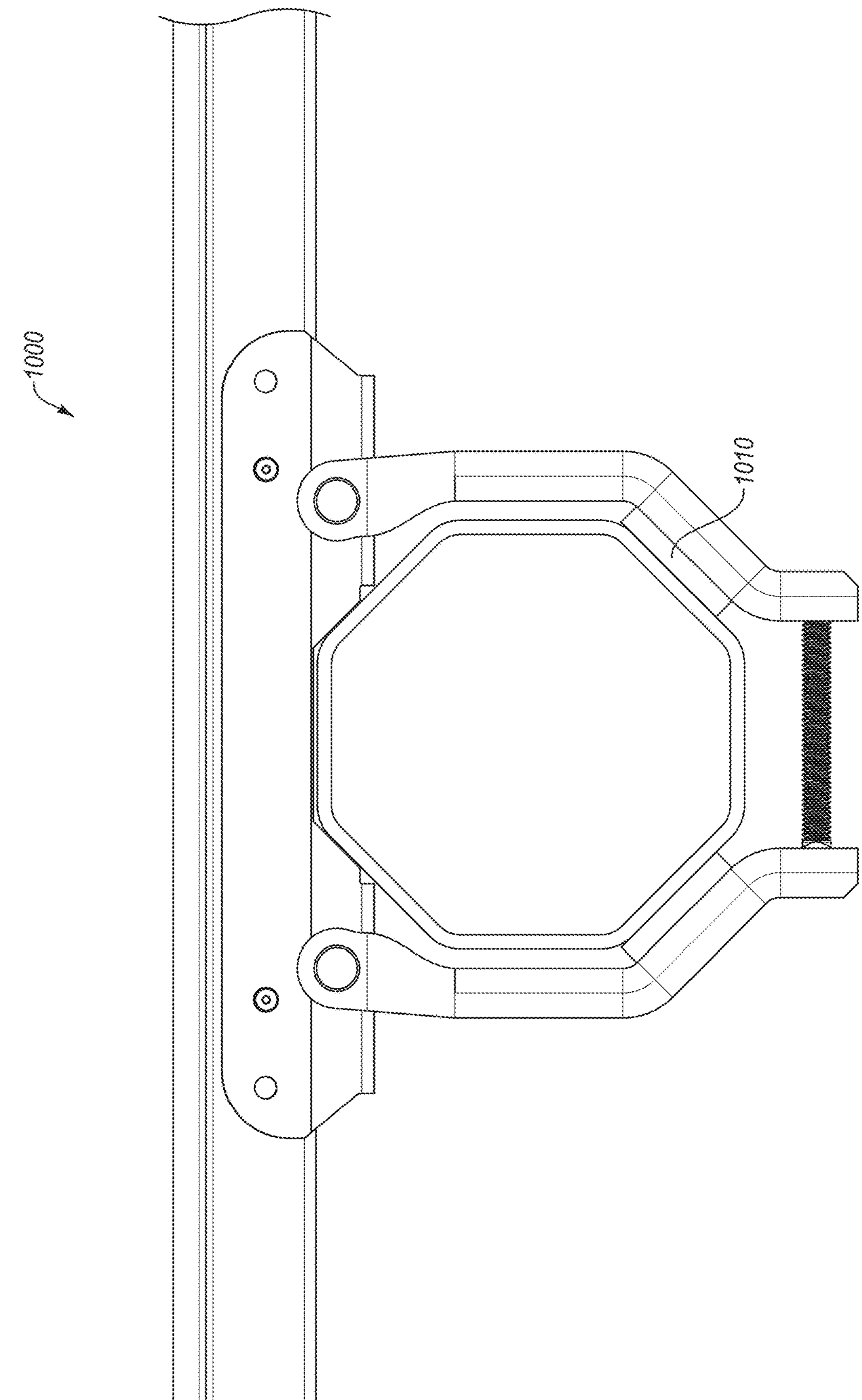

FIG. 10B

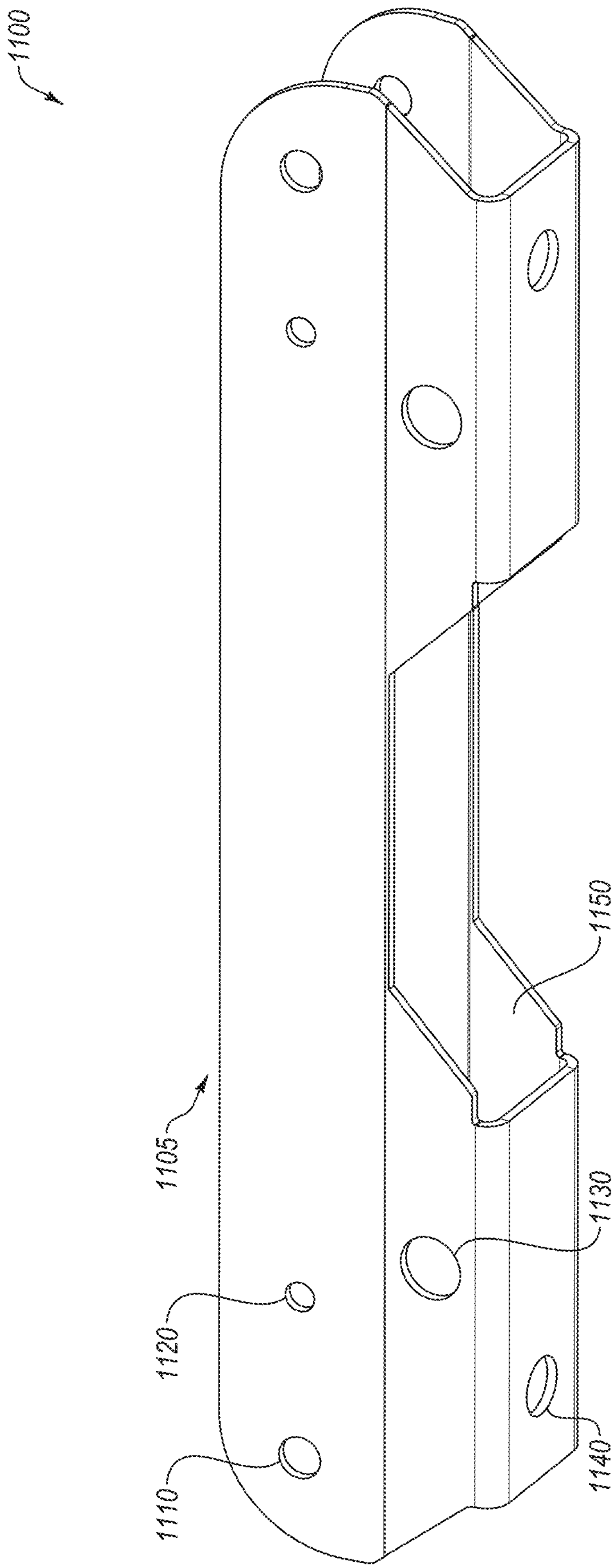

FIG. 11

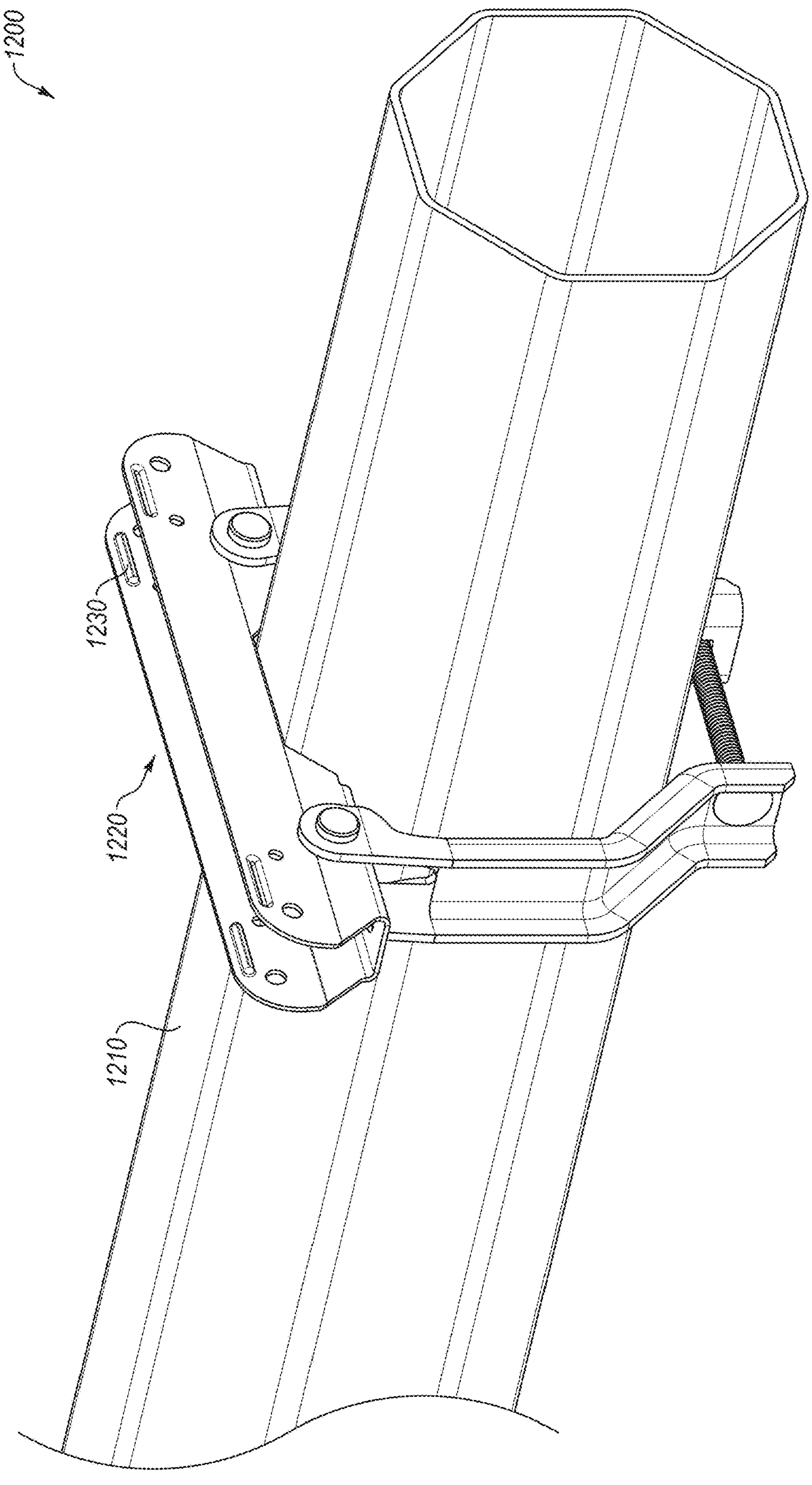

FIG. 12A

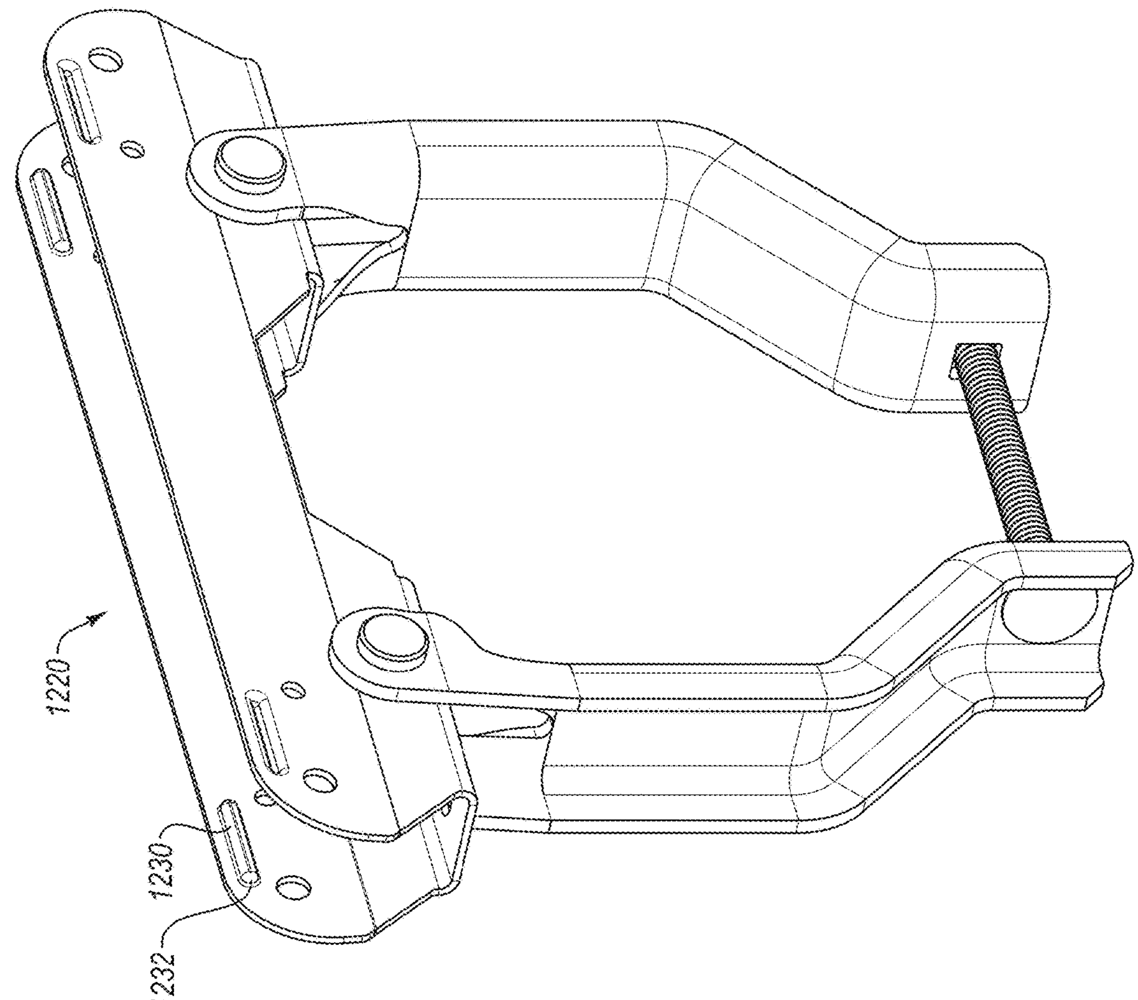

FIG. 12B

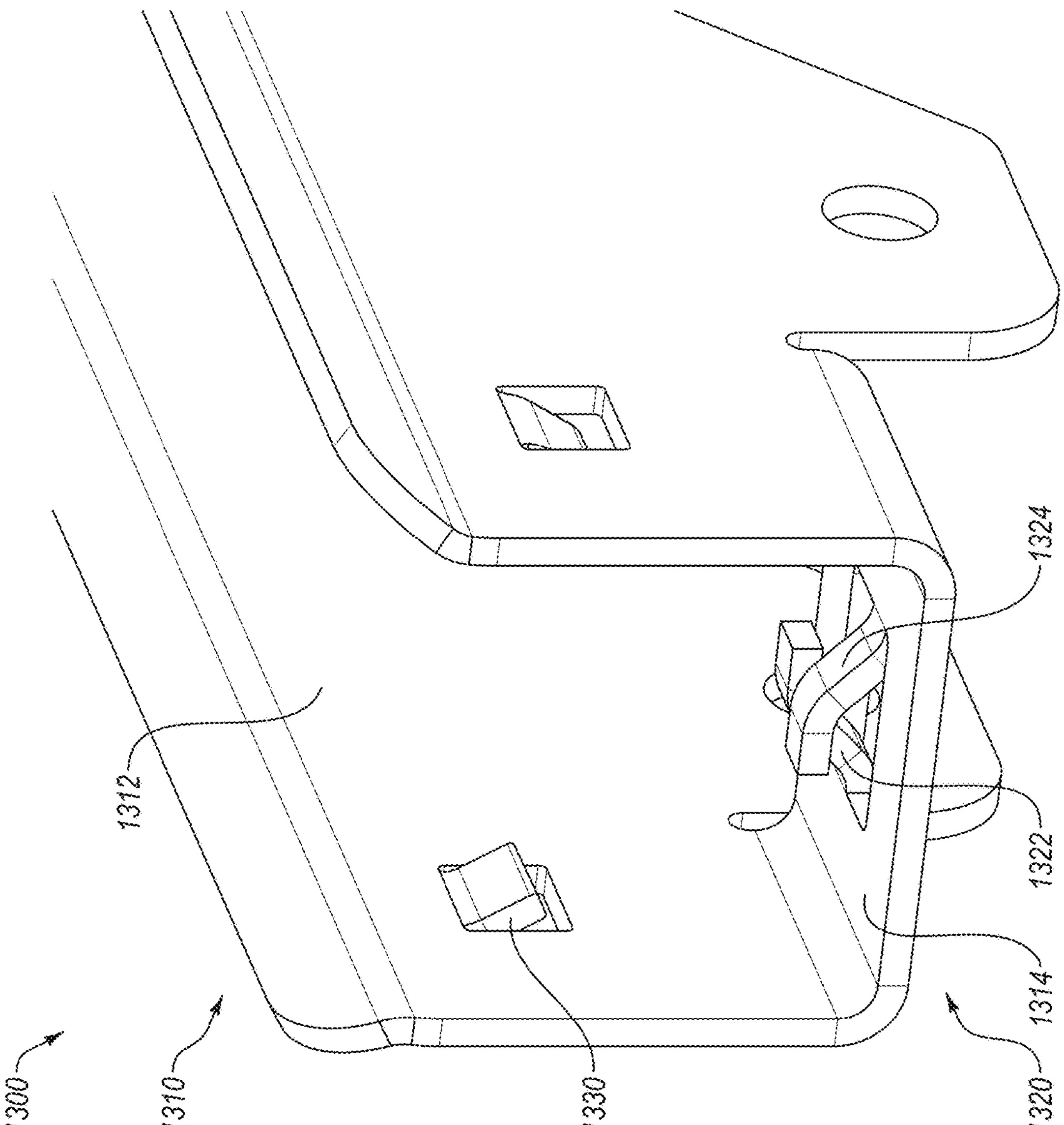

FIG. 13

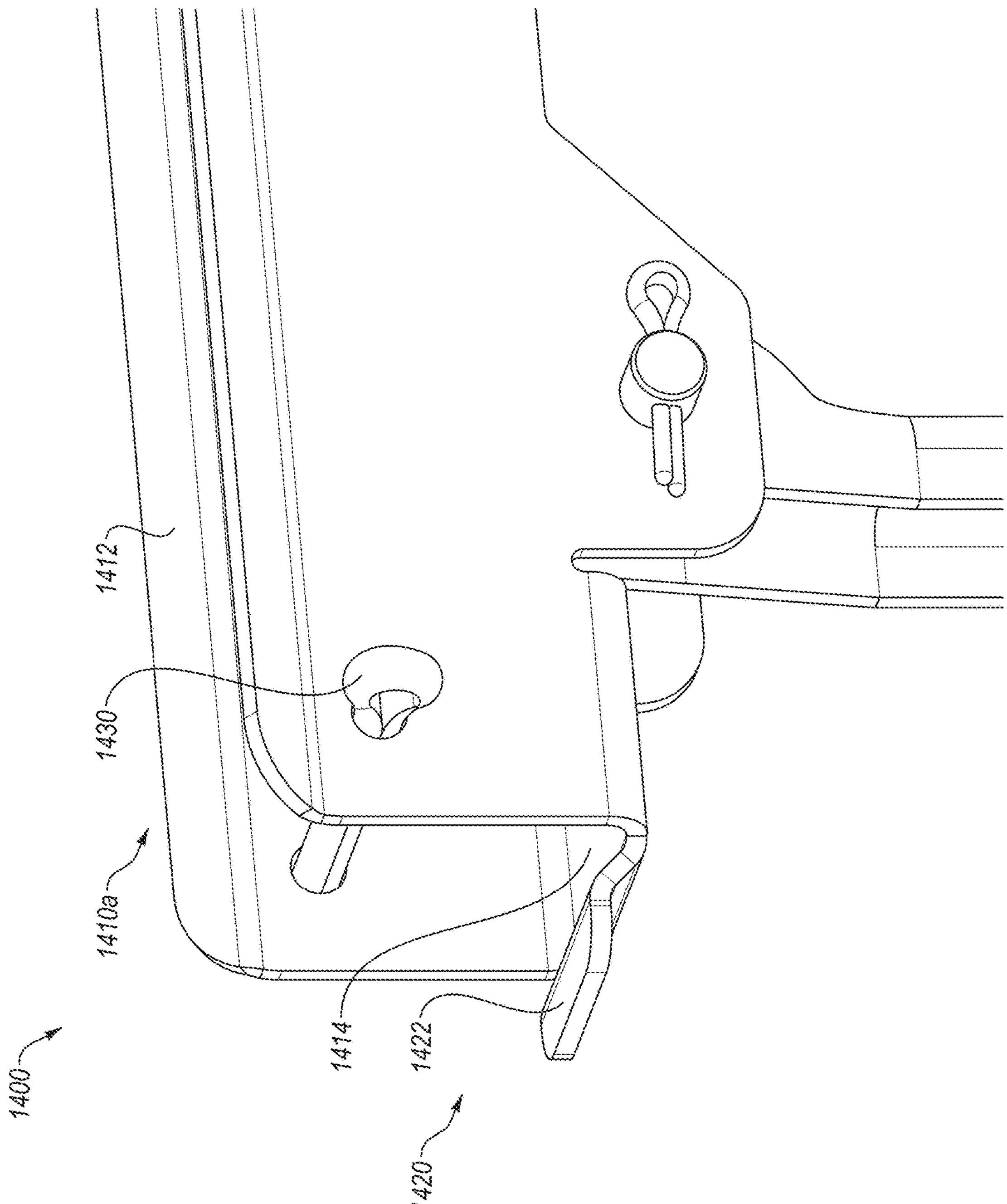

FIG. 14A

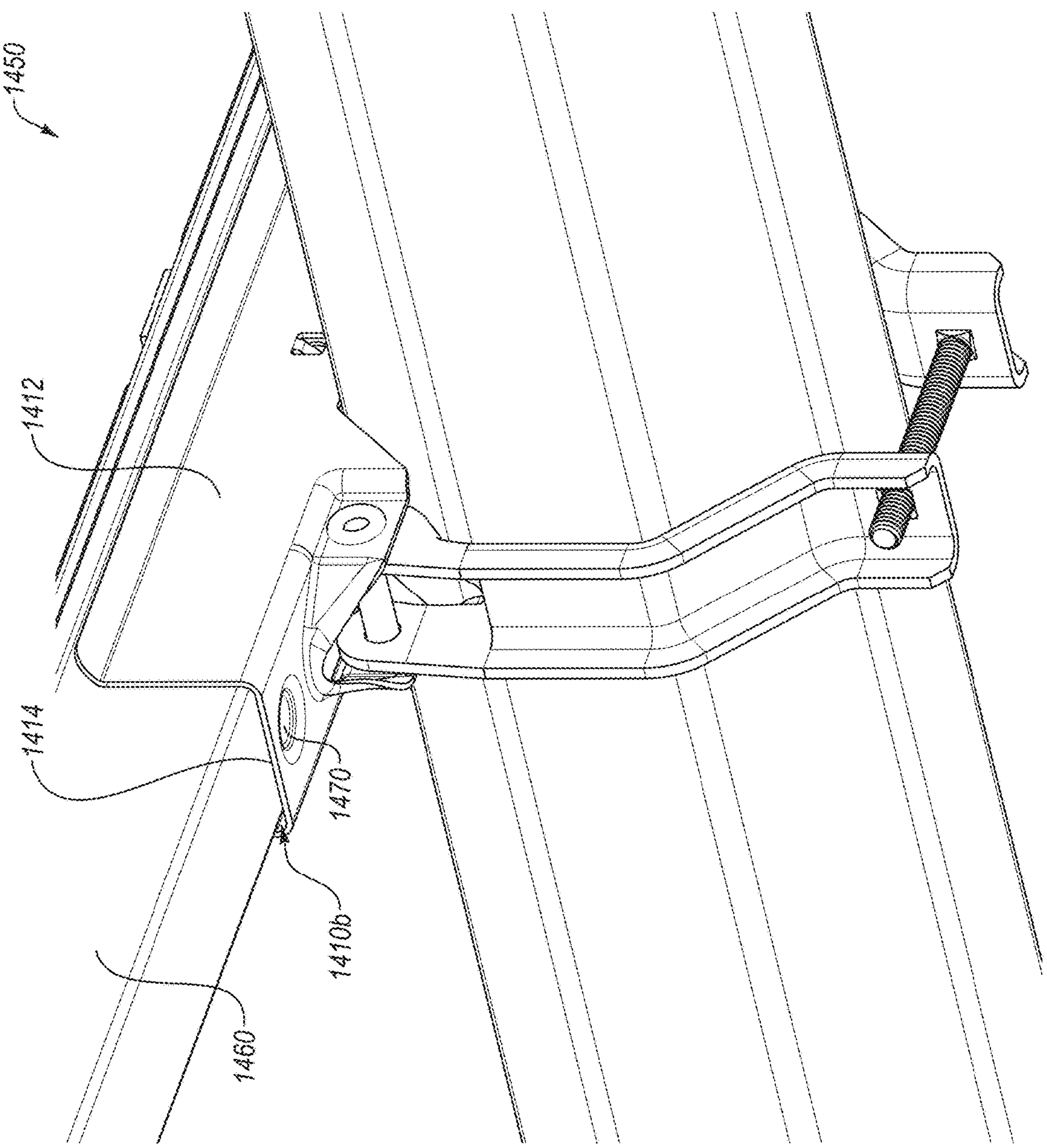

FIG. 14B

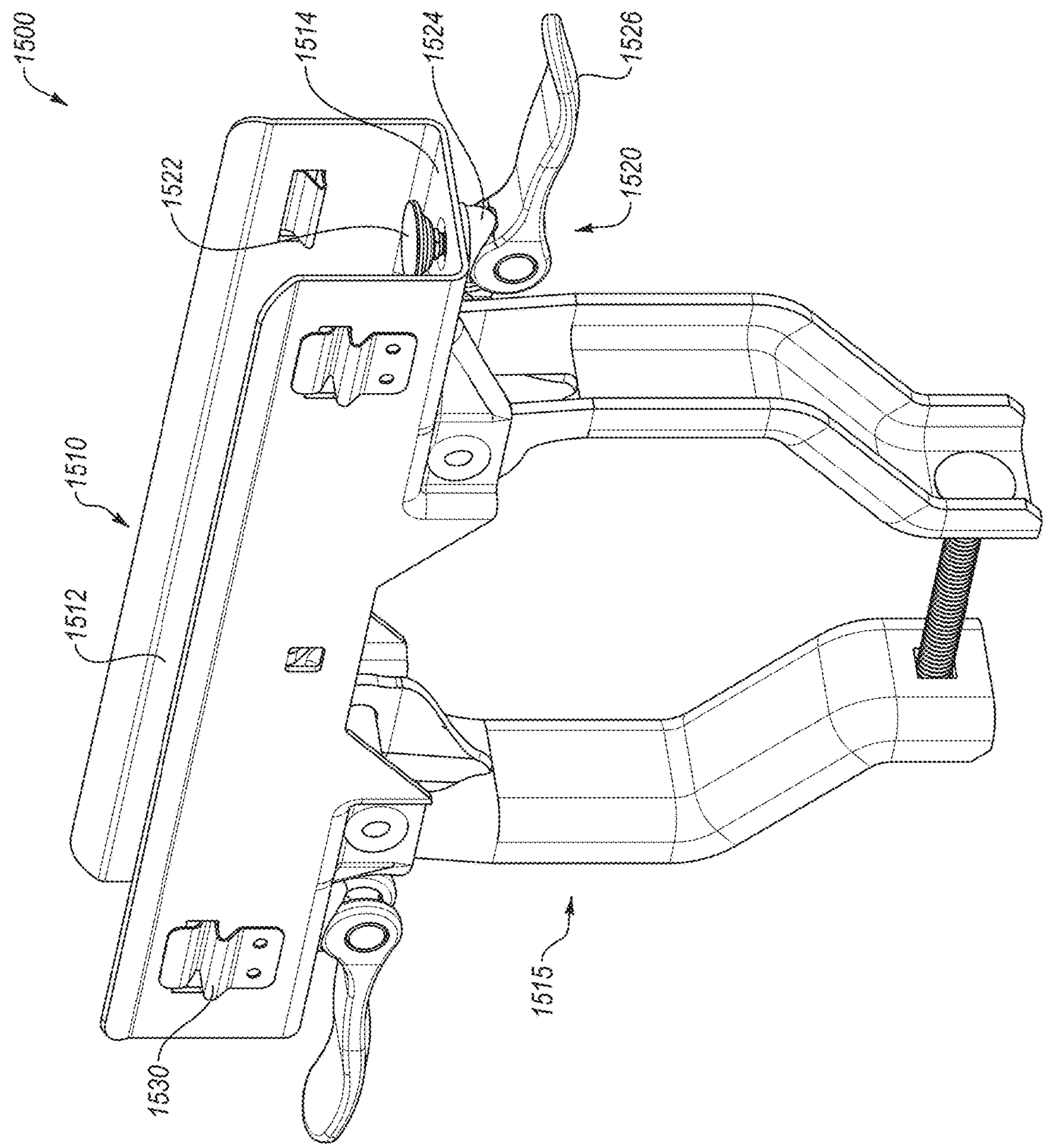

FIG. 15

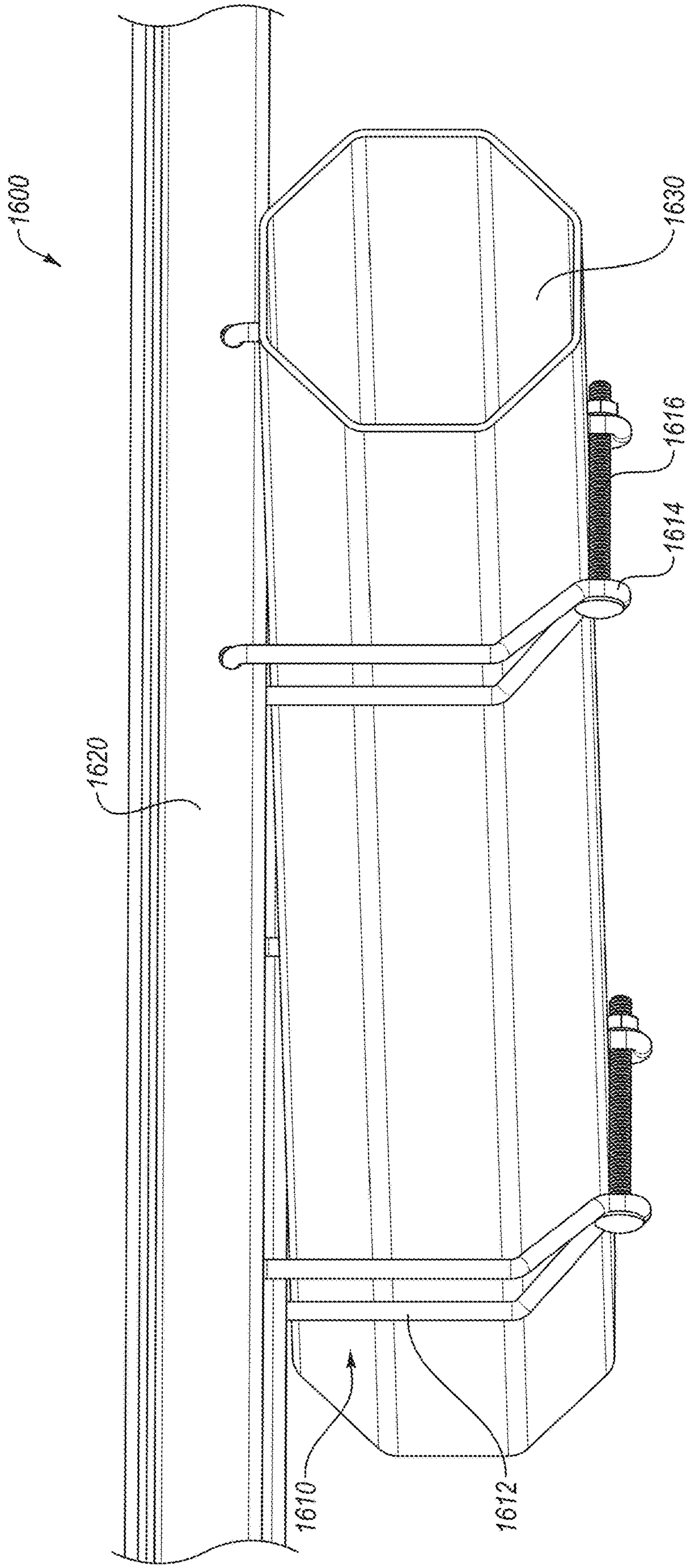

FIG. 16A

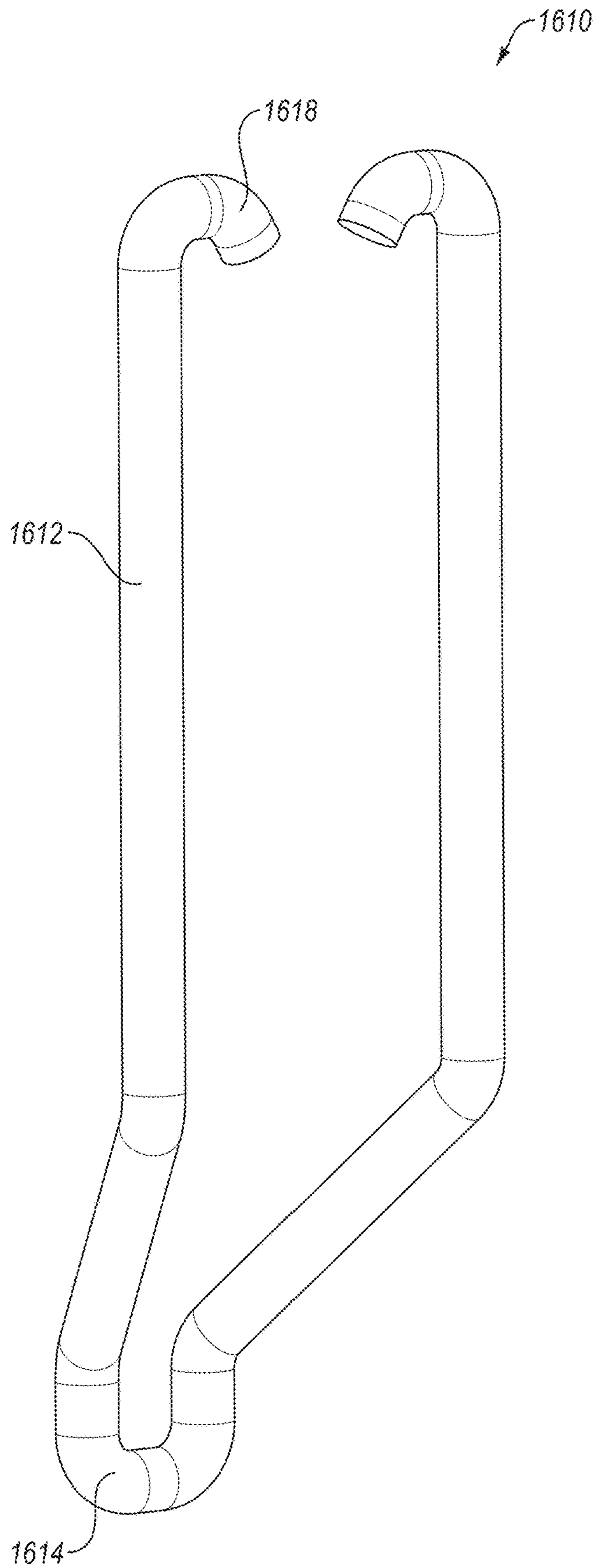

FIG. 16B

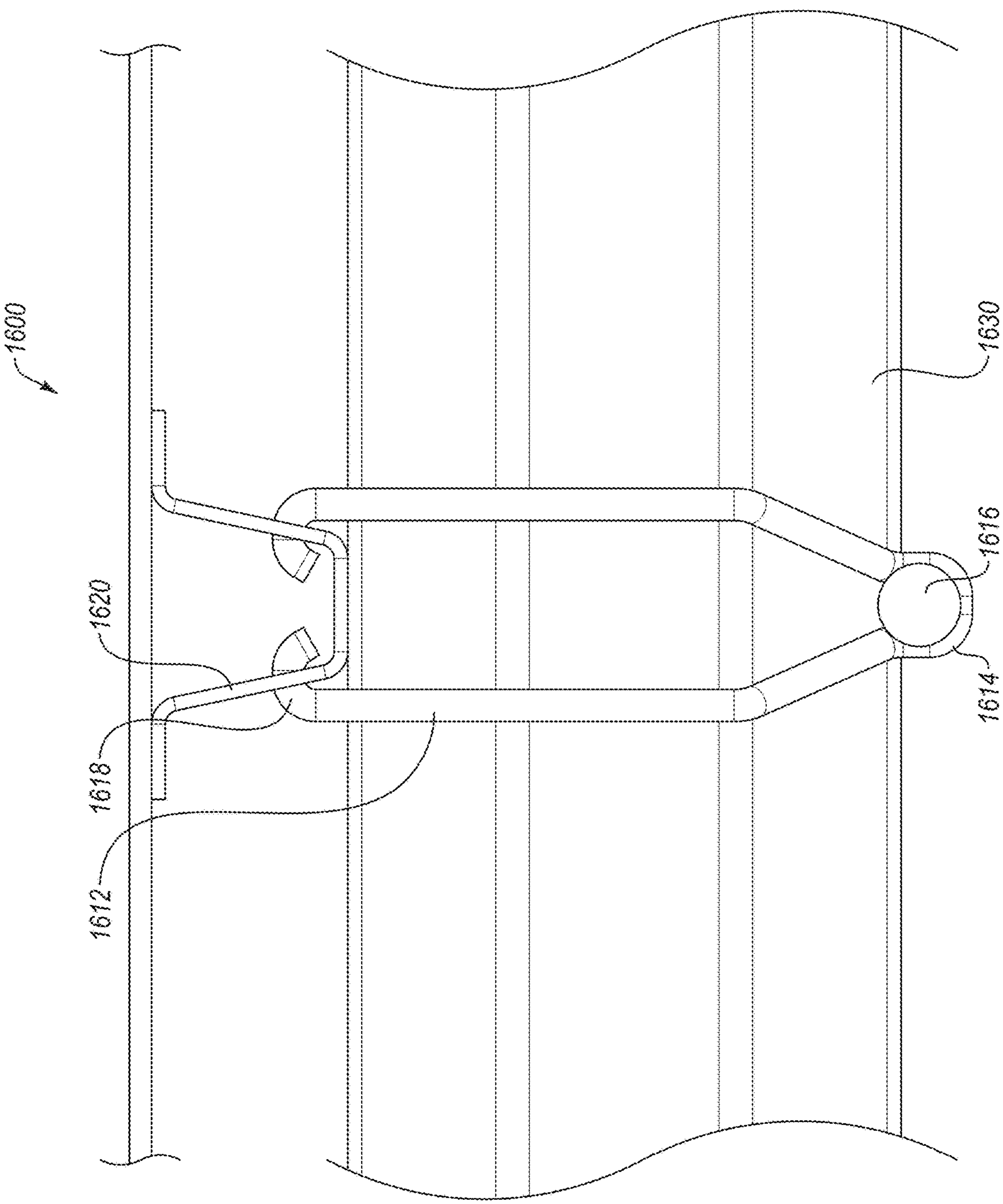

FIG. 16C

SNAP-LOCK PHOTOVOLTAIC MODULE MOUNTING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit of U.S. patent application Ser. No. 17/652,440, filed on Feb. 24, 2022, which claims the benefit of and priority to U.S. Patent Application Ser. No. 63/228,261, filed on Aug. 2, 2021, and U.S. Patent Application Ser. No. 63/153,326, filed on Feb. 24, 2021; the disclosures of which are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

The present disclosure generally relates to a snap-lock photovoltaic module mounting system.

SUMMARY OF THE INVENTION

According to an aspect of the present disclosure, a photovoltaic (PV) module clamp may be configured to interface with a torque tube and a PV module rail fixedly coupled to a PV module. The PV module clamp may include a seating portion made of two or more lateral walls and a base surface. A clamp body of the PV module clamp may be coupled to the seating portion. The clamp body may include a shape corresponding to a cross-sectional shape of the torque tube. The PV module clamp may include a fastening feature for interlocking the PV module clamp with the PV module rail responsive to the PV module rail being seated in the seating portion of the PV module clamp.

The fastening feature of the PV module clamp of any of the foregoing examples may additionally or alternatively include at least one of: a nub, an insertion peg, a spring tab, a hook tab, and a spring-loaded protrusion positioned along an interior surface of one or more of the lateral walls of the PV module clamp and positioned to interlock with a corresponding hole in the PV module rail. The fastening feature of the PV module clamp of any of the foregoing examples may additionally or alternatively include an external wire form configured to rotate and interface with at least one of the PV module clamp or the PV module rail.

One or more of the lateral walls of the PV module clamp of any of the foregoing examples may additionally or alternatively include a tab upon which the fastening feature is disposed, the tab shaped to flex outwards when the PV module rail is passing the fastening feature while being biased back to its original position such that when a hole in the PV module rail aligns with the fastening feature, the tab may return to its original position interlocking the fastening feature in the hole in the PV module rail. One or more of the lateral walls of the PV module clamp of any of the foregoing examples may additionally or alternatively flex outwards when the PV module rail is passing the fastening feature while being biased back to its original position such that when a hole in the PV module rail aligns with the fastening feature, the at least one of the lateral walls may return to their original position interlocking the fastening feature in the hole in the PV module rail.

The shape of the clamp body of the PV module clamp of any of the foregoing examples may additionally or alternatively allow the clamp body to at least partially circumscribe the torque tube.

The base surface of the seating portion of the PV module clamp of any of the foregoing examples may additionally or alternatively include one or more bridge lances that protrude into an interior space of the seating portion. The base surface of the seating portion of the PV module clamp of any of the foregoing examples may additionally or alternatively include an extended tab that extends away from an interior space of the seating portion.

The PV module clamp of any of the foregoing examples may additionally or alternatively include a pressure pad made of a pad component and a body component, the pressure pad protruding from the base surface of the seating portion and being configured to interface with the PV module rail such that a gap between the PV module rail and the base surface of the seating portion is formed. The pressure pad of any of the foregoing examples may additionally or alternatively include a cam lever that is configured to adjust a height of the pressure pad. The body component of the pressure pad of any of the foregoing examples may additionally or alternatively extend through the cam lever such that rotation of the cam lever or the body component adjusts the height of the pressure pad.

According to an aspect of the present disclosure, a PV module system may include a PV module rail coupled to a PV module, a torque tube, and a PV module clamp configured to interface with the torque tube and the PV module rail. The PV module clamp may include a seating portion made of two or more lateral walls and a base surface. A clamp body of the PV module clamp may be coupled to the seating portion. The clamp body may include a shape corresponding to a cross-sectional shape of the torque tube. The PV module clamp may include a fastening feature for interlocking the PV module clamp with the PV module rail responsive to the PV module rail being seated in the seating portion of the PV module clamp.

The fastening feature of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include at least one of: a nub, an insertion peg, a spring tab, a hook tab, and a spring-loaded protrusion positioned along an interior surface of one or more of the lateral walls of the PV module clamp and positioned to interlock with a corresponding hole in the PV module rail. The fastening feature of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include an external wire form configured to rotate and interface with at least one of the PV module clamp or the PV module rail.

One or more of the lateral walls of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include a tab upon which the fastening feature is disposed, the tab shaped to flex outwards when the PV module rail is passing the fastening feature while being biased back to its original position such that when a hole in the PV module rail aligns with the fastening feature, the tab may return to its original position interlocking the fastening feature in the hole in the PV module rail. One or more of the lateral walls of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively flex outwards when the PV module rail is passing the fastening feature while being biased back to its original position such that when a hole in the PV module rail aligns with the fastening feature, the at least one of the lateral walls may return to their original position interlocking the fastening feature in the hole in the PV module rail.

The shape of the clamp body of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively allow the clamp body to at least partially circumscribe the torque tube.

The base surface of the seating portion of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include one or more bridge lances that protrude into an interior space of the seating portion. The base surface of the seating portion of the PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include an extended tab that extends away from an interior space of the seating portion.

The PV module clamp of the PV module system of any of the foregoing examples may additionally or alternatively include a pressure pad made of a pad component and a body component, the pressure pad protruding from the base surface of the seating portion and being configured to interface with the PV module rail such that a gap between the PV module rail and the base surface of the seating portion is formed. The pressure pad of any of the foregoing examples may additionally or alternatively include a cam lever that is configured to adjust a height of the pressure pad. The body component of the pressure pad of any of the foregoing examples may additionally or alternatively extend through the cam lever such that rotation of the cam lever or the body component adjusts the height of the pressure pad.

The object and advantages of the embodiments will be realized and achieved at least by the elements, features, and combinations particularly pointed out in the claims. It is to be understood that both the foregoing general description and the following detailed description are explanatory and are not restrictive of the invention, as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

Example embodiments will be described and explained with additional specificity and detail through the accompanying drawings in which:

FIG. **1**A is an isometric view of an example embodiment of a snap-lock PV module mounting system including a PV module mounted on top of one or more module clamps according to at least one embodiment of the present disclosure;

FIG. **1**B is a side view of the example embodiment of the snap-lock PV module mounting system;

FIG. **1**C is the side view of the example embodiment of the snap-lock PV module mounting system including the PV module mounted on top of the module clamps and a spring clip according to at least one embodiment of the present disclosure;

FIG. **2** is a diagram of a spring clip that may be installed within a PV module frame according to at least one embodiment of the present disclosure;

FIG. **3**A is an isometric view a first example embodiment of a module clamp of the snap-lock PV module mounting system including a seating portion with protruding nubs according to at least one embodiment of the present disclosure;

FIG. **3**B is a side view of the first example embodiment of the module clamp;

FIG. **3**C is an end view of the first example embodiment of the module clamp;

FIG. **4**A is an isometric view of a second example embodiment of a module clamp with insertion pegs according to at least one embodiment of the present disclosure;

FIG. **4**B is a side view of the second example embodiment of the module clamp;

FIG. **4**C is an end view of the second example embodiment of the module clamp;

FIG. **5**A is an isometric view of a third example embodiment of a module clamp with external wire forms according to at least one embodiment of the present disclosure;

FIG. **5**B is a side view of the third example embodiment of the module clamp;

FIG. **5**C is an end view of the third example embodiment of the module clamp;

FIG. **5**D is a side view of the third example embodiment of the module clamp including a second configuration of an external wire form positioned to secure a module rail;

FIG. **5**E is a side view of the third example embodiment of the module clamp including a third configuration of the external wire form positioned to secure the module rail;

FIG. **6** is an isometric view of a fourth example embodiment of the module clamp with spring tabs according to at least one embodiment of the present disclosure;

FIG. **7**A is an isometric view of a fifth example embodiment of the module clamp with hook tabs according to at least one embodiment of the present disclosure;

FIG. **7**B is a side view of the fifth example embodiment of the module clamp;

FIG. **7**C is a cross-sectional view of the fifth example embodiment of the module clamp;

FIG. **8** is an isometric view of a sixth example embodiment of the module clamp with a spring-loaded protrusion according to at least one embodiment of the present disclosure;

FIG. **9** illustrates a side view of a custom nut according to at least one embodiment of the present disclosure;

FIG. **10**A is an isometric view of a seventh example embodiment of the module clamp according to at least one embodiment of the present disclosure;

FIG. **10**B is a side view of the seventh example embodiment of the module clamp;

FIG. **11** illustrates a seating portion of the module clamp according to at least one embodiment of the present disclosure;

FIG. **12**A is an isometric view of an eighth example embodiment of the module clamp coupled to a torque tube according to at least one embodiment of the present disclosure;

FIG. **12**B is an isometric view of the eighth example embodiment of the module clamp according to at least one embodiment of the present disclosure;

FIG. **13** is a close-up view of a ninth example embodiment of the module clamp according to at least one embodiment of the present disclosure;

FIGS. **14**A and **14**B are close-up views of a tenth and an eleventh example embodiment of the module clamp according to at least one embodiment of the present disclosure;

FIG. **15** is an isometric view of a twelfth example embodiment of the module clamp according to at least one embodiment of the present disclosure;

FIG. **16**A is an isometric view of a thirteenth example embodiment of the module clamp coupled to a torque tube according to at least one embodiment of the present disclosure;

FIG. **16**B illustrates a wire form clamp body of the thirteenth embodiment of the module clamp according to at least one embodiment of the present disclosure; and

FIG. **16**C is a side view of the module clamp coupled to the torque tube according to at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

The present disclosure relates to the mounting of PV modules to support structure, such as that used to facilitate tracking of the sun via the PV modules. For example, PV modules may be mounted through a variety of hardware or other structural components to a torque tube or other device for varying the orientation of the PV modules to track the sun as it progresses across the sky. In some circumstances, the PV modules may be mounted to one or more rails running along a back side of the PV module(s).

FIGS. **1**A and **1**B illustrate different views of an example embodiment of a snap-lock PV module mounting system **100**. FIG. **1**A is an isometric view of an example embodiment of a snap-lock PV module mounting system **100** including a PV module **110** mounted on top of one or more module clamps **140** according to at least one embodiment of the present disclosure. FIG. **1**B is a side view of the example embodiment of the snap-lock PV module mounting system **100**, and FIG. **1**C is the side view of the example embodiment of the snap-lock PV module mounting system **100** including the PV module **110** mounted on top of the module clamps **140**.

The snap-lock PV module mounting system **100** may include a PV module **110** that includes one or more module rails **120**. Each of the module rails **120** may interface with one of the module clamps **140** such that the module rails **120** are attached and/or fixed to the module clamps **140**. The module clamps **140** may circumscribe a module support structure **130** (e.g., a torque tube as illustrated) or attach to the module support structure **130** by another method such that the PV module **110** is coupled to the module support structure **130**.

In some embodiments, the module clamps **140** may include a clamp band **142**, a seating portion **144**, and/or one or more protruding features **146**. In some embodiments, the protruding features **146** may be positioned to extend from an interior surface of the seating portion **144**, such as from lateral walls **148** of the seating portion **144**. In some embodiments, the module rail **120** may be positioned over the module clamps **140** and set down into the seating portion **144** of the module clamps **140** during installation such that the module rail **120** is seated within the seating portion **144**. Additionally or alternatively, the module rail **120** may be positioned in any way such that the module rail **120** may interface with the seating portion **144** of the module clamp **140**. For example, the module rail **120** may be oriented in a horizontal (or any other) direction, such as by propping up the PV module **110**, and the module rail **120** may be slotted into the seating portion **144** of the module clamp **140**. As another example, the module rail may be oriented in an upward direction, and the seating portion **144** may be aligned and set onto the module rail **120** from above.

In some embodiments, one or more lateral walls of the seating portion **144** may be angled such that placing the module rail **120** in the seating portion **144** and/or placing the seating portion **144** onto the module rail **120** pushes or flexes the lateral walls of the seating portion **144** outward as the module rail is pressed downwards into the seating portion **144** towards the module support structure **130**. In some embodiments, the module rail **120** may include one or more openings into which the protruding features **146** may interface after the module rail **120** has been pushed a sufficient distance into the seating portion **144**. The protruding features **146** interfacing with the openings of the module rail **120** may cause or allow the lateral walls of the seating portion **144** to contract back towards their original positions and lock the module rail **120** in place, which may fix the position of the PV module **110** relative to the module support structure **130**. For example, such pushing outward or flexion of the lateral walls of the seating portion **144** outward may be an elastic deformation of the lateral walls of the seating portion **144** such that they may be biased towards their original position and may move back towards their original position when the protruding features **146** align with holes in the module rail **120**.

In some embodiments, the seating portion **144** may be sized and/or positioned and/or the protruding features **146** may be positioned such that the module rail **120** is positioned against the module support structure **130** when fully seated in the seating portion **144**.

In some embodiments, a spring clip **200** may be used in conjunction with the module clamp **140**. FIG. **1**C is the side view of the example embodiment of the snap-lock PV module mounting system **100** including the PV module **110** mounted on top of the module clamp **140** in which the module clamp **140** includes the spring clip **200** according to at least one embodiment of the present disclosure. FIG. **2** is a diagram of a spring clip **200** that may be installed within a PV module frame and/or a module rail (such as the module rail **120**) according to at least one embodiment of the present disclosure. The spring clip **200** may include one or more protruding features **210** coupled to a clip band **220**. In some embodiments, the spring clip **200** may be attached to or disposed within the PV module frame and/or the module rail with the protruding features **210** on the spring clip **200** extending beyond the PV module frame and/or module rail.

In these and other embodiments, the clip band **220** may be compressed as the PV module frame/rail is positioned against a module clamp (e.g., seated within the seating portion **144** of the module clamp **140**). In these and other embodiments, the lateral walls of the module clamp may include one or more openings corresponding to the position and/or size of the protruding features **210** of the spring clip **200**. As such, the clip band **220** may be compressed until the protruding features **210** are aligned and interface with the openings in the lateral walls of the module clamp, in which circumstance the spring force of the clip band **220** from being compressed may cause the protruding features **210** to spring back outward into the holes. Such an interaction may lock the position of the spring clip **200** relative to the module clamp, thereby also locking the PV module frame/rail relative to the module clamp.

FIG. **3**A is an isometric view of an example embodiment of a module clamp **300** including a seating portion **320** with protruding nubs **330** according to at least one embodiment of the present disclosure. FIG. **3**B is a side view of the second example embodiment of the module clamp **300**, and FIG. **3**C is an end view of the second example embodiment of the module clamp **300**. The embodiments illustrated in FIGS. **3**A-**3**C may operate in a similar manner to those described in reference to FIGS. **1**A-**1**C, with a module rail being pressed into the module clamp to seat the module rail in the seating portion to lock the module rail (and the PV modules attached thereto) in place relative to the module clamp.

In some embodiments, the protruding nubs **330** may be a fastening feature made of a metal, such as cast iron, stainless steel, titanium, or aluminum, among others, and shaped by a sheet metal stamping process. For example, a circular, semicircular, or other shape may be stamped or punched into a sheet of metal used to create the seating area of the seating portion **320** with sufficient force to indent portions of the stamped shape inwards (e.g., such that the protruding nubs **330** are projecting inwards into the seating area of the seating portion **320**) while keeping a portion of the shape still attached to the seating area of the seating portion **320**.

The seating portion **320** of the module clamp **300** may be coupled to a clamp body **310** that is configured to interface with a torque tube. In some embodiments, the clamp body **310** may be made of one or more components that are shaped based on a shape of the torque tube with which the clamp body **310** is configured to interface. For example, the example of the clamp body **310** corresponding to a torque tube having an octangular cross-section illustrated in FIG. **3**A may include a first body section **312** that corresponds to a straight edge of the octangular torque tube and a second body section **314** connected to the first body section **312** that corresponds to an angled edge of the octangular torque tube. In this and other examples, the example clamp body **310** may include a third body section **316** that is connected to the second body section **314** and is parallel to the first body section **312**. A bolt, screw, or any other fastener **340** may be inserted through the third body sections **316** of two or more adjacent clamp body components to connect the clamp body components together via their respective third body sections **316** as illustrated in FIG. **3**A. While illustrated as being parallel, it will be appreciated that the third portion may extend in any direction such that it does not interfere with the torque tube and provides a surface against which the fastener **340** may be tightened.

In some embodiments, a profile of the seating portion **320** may include a base surface **322** and two lateral walls **324**. The base surface **322** may be a bottom of the seating portion **320** against which the module rail may be pressed when it is fully seated in the seating portion **320**. Additionally or alternatively, when seated in the seating portion **320**, the module rail may be some distance away from the base surface **322** despite being fully seated. In these and other embodiments, the base surface **322** may act as a stabilizing and connecting component between the two lateral walls **324**. In some embodiments, the lateral walls **324** may include a generally vertical portion and a flared portion. The flared portion may make it easier for an installer to guide the module rail into the seating portion **320**. Additionally or alternatively, the flared portion may facilitate flexion of the lateral walls **324** outwards when pushed outwards by the module rail interfering with the protruding nubs **330**. In these and other embodiments, when the protruding nubs **330** are disposed on the flared portion, the protruding features may extend into the interior region **326** of the seating portion **320** (e.g., the protruding nub **330** may extend beyond the generally vertical portions of the lateral walls **324**).

FIG. **4**A is an isometric view of a second example embodiment of a module clamp **400** with insertion pegs **430** according to at least one embodiment of the present disclosure. FIG. **4**B is a side view of the third example embodiment of the module clamp **400**, and FIG. **4**C is an end view of the third example embodiment of the module clamp **400**. The embodiments illustrated in FIGS. **4**A-**4**C may operate in a similar manner to those described in reference to FIGS. **1**A-**1**C, with a module rail being pressed into the module clamp to seat the module rail in the seating portion to lock the module rail (and the PV modules attached thereto) in place relative to the module clamp.

In some embodiments, the insertion pegs **430** may be fastening features that are punched, bored, drilled, or otherwise inserted through one or more lateral walls of the seating portion. Additionally or alternatively, one or more openings may be punched, bored, drilled, cast, or otherwise formed in or through the lateral walls of the seating portion, and the insertion pegs **430** may be inserted through the openings. For example, openings may be drilled through the lateral walls of the seating portion, and the insertion pegs **430** may include pop rivets or other rivets that are inserted and locked into the openings. As another example, the insertion pegs **430** may include threaded or partially threaded pegs that are drilled into the lateral walls of the seating portion via a flow drill fastening process. As an additional example, the insertion pegs **430** may include a bolt and/or nut that is sized and positioned to correspond to openings in the PV module frame/rail.

FIG. **5**A is an isometric view of a third example embodiment of a module clamp **500** with external wire forms **530** according to at least one embodiment of the present disclosure. FIG. **5**B is a side view of the fourth example embodiment of the module clamp **500**, and FIG. **5**C is an end view of the fourth example embodiment of the module clamp **500**. The embodiments illustrated in FIGS. **5**A-**5**C may operate in a similar manner to those described in reference to FIGS. **1**A-**1**C, with a module rail being pressed into the module clamp to seat the module rail in the seating portion to lock the module rail (and the PV modules attached thereto) in place relative to the module clamp.

In some embodiments, the external wire forms **530** may be fastening features that include a wire that is bent, cast, or otherwise formed to interface with one or more openings in the lateral walls **524** of the seating portion. For example, the external wire forms **530** may include a wire bent into a rounded rectangular shape with an open edge (as illustrated in FIGS. **5**A-**5**C), a triangular shape, a curved or hemispherical shape, or any other suitable shape. A first end and a second end of the external wire form **530** may each interface with an opening in the lateral walls **524** of the seating portion **520** such that the external wire form **530** is coupled to the module clamp. In these and other embodiments, the ends of the external wire form **530** may extend through the openings in the lateral walls **524** and into the interior region of the seating portion and function in the same or a similar manner as the protruding nubs **330** and/or the insertion pegs **430** as described above in relation to FIGS. **3**A-**3**C and/or **4**A-**4**C, respectively. For example, as a module rail is pressed into the seating portion **520**, the module rail may interfere with the ends of the external wire form **530** and force the external wire form **530** to deform in such a way that the ends are pushed back out of their holes far enough for the module rail to pass the ends of the external wire form **530** towards the base surface **522** and be seated in the seating portion **520**. An opening of the module rail may align with the ends of the external wire form **530** when fully seated such that when aligned, the external wire form **530** springs back to its original position when the ends of the external wire form **530** are able to pop into the openings in the module rail.

In addition to the locking in place obtained by the ends of the external wire form **530** popping into the opening in the module rail, the external wire form **530** may be used to provide an additional locking feature. For example, the external wire forms **530** may include an elongated length such that the external wire forms **530** may rotate underneath the seating portions of the module clamps and further secure module rails and/or PV modules coupled to the module clamps (for example, as illustrated in a snap-lock PV module mounting system **500***d* shown in FIG. **5**D). For example, the external wire form **530** may be rotated past a ridge **526** over which the external wire form **530** is to deform to get past. On the other side of the ridge **526**, the external wire form **530** may reside in a valley **528** that may keep the external wire form **530** in some tension so that the ends of the external wire form **530** are less likely to pop out and/or otherwise may more securely retain the external wire form **530** in place. Additionally or alternatively, the external wire forms **530** may be shaped or formed such that the external wire form **530** is configured to rotate up and around a portion of a module rail **510** (for example, as illustrated in a snap-lock PV module mounting system **500***e* shown in FIG. **5**E). For example, the external wire form **530** may extend over a ridge **526** in the module rail and into a valley **528** on the other side of the ridge **526** such that without significant external force, the external wire form **530** may remain in the valley **528** and will not traverse up over the ridge **526**. By doing so, the module rail **510** may be more securely affixed to the module clamp of the snap-lock PV module mounting system **500***e*.

FIG. **6** is an isometric view of a fourth example embodiment of a module clamp **600** with spring tabs **630** according to at least one embodiment of the present disclosure. The embodiments illustrated in FIGS. **6**A-**6**C may operate in a similar manner to those described in reference to FIGS. **1**A-**1**C, with a module rail being pressed into the module clamp to seat the module rail in the seating portion to lock the module rail (and the PV modules attached thereto) in place relative to the module clamp.

The spring tabs **630** may be fastening features that include one or more sections cut out from the lateral walls of the seating portion such that the cut-out sections are capable of some degree of movement via flexion at the interface points of the spring tabs **630** and are attached to the lateral walls of the seating portion on at least one edge. In some embodiments, the spring tabs **630** may be oriented such that the cut-out sections face the open end of the seating portion (as illustrated in FIG. **6**). Additionally or alternatively, the spring tabs **630** may be oriented such that the cut-out sections face a base surface of the seating portion. In these and other embodiments, placing the module rail in the seating portion may push the lateral walls of the module clamp outward and facilitate the spring tabs **630** interfacing with one or more corresponding openings, hooks, and/or any other structures of the module rail. Additionally or alternatively, the module rail and/or the seating portion may be sized to be comparable in size such that the lateral walls of the seating portion may not be displaced by the module rail being pressed into the seating portion but may push the spring tabs **630** outward until openings in the module rail are aligned with the protrusions on the spring tabs **630**. When aligned, the spring force caused by the outwards flexion of the spring tabs **630** may cause the protrusions of the spring tabs **630** to spring into the holes of the module rail.

FIG. **7**A is an isometric view of an example embodiment of a module clip **750** with hook tabs **730** according to at least one embodiment of the present disclosure. FIG. **7**B is a side view of the fifth example embodiment of the module clip **750**, and FIG. **7**C is a cross-sectional view of a fifth example embodiment of a module clamp **700**. In some embodiments, the hook tabs **730** may be fastening features that interface with one or more openings of a module rail **710** as the module rail **710** is seated in a seating portion **720**. A flange **735** at the end of each of the hook tabs **730** may prevent the module rail **710** and module clamp **700** from disengaging from one another.

In these and other embodiments, the module rail **710** may be seated into the seating portion **720** of a module clamp **700** (such as shown in FIG. **7**B by the module rail being flush against the flared portions of the seating portion **720**), followed by the module clip **750** being forced from below up and over the module clamp **700** until the hook tabs **730** are forced outwards by the flaring of the seating portion **720**. When the flange **735** of the module clip **750** align with holes in the seating portion **720** and/or the module rail **710**, the spring force from deforming the module clip **750** may cause the flanges **735** to extend into the holes.

In the embodiment illustrated in FIG. **7**C, a similar hook tab **730** with flanges **735** as that illustrated in FIGS. **7**A and **7**B may be used in a similar or comparable manner as the protruding feature of FIGS. **1**A-**1**C. For example, lateral walls **722** may flare or flex outwards as the module rail **710** is pressed into the seating portion **720** until holes in the module rail **710** align with the hook tabs **730** and the hook tabs **730** spring into the aligned holes to lock the module rail **710** in place relative to the seating portion **720**.

FIG. **8** is an isometric view of a sixth example embodiment of a module clamp **800** with a spring-loaded protrusion **830** according to at least one embodiment of the present disclosure. The spring-loaded protrusion **830** may be a fastening feature that includes a peg **832** projecting into the interior region of a seating portion **810** (shown as the dashed box corresponding to the peg **832**, indicating it is seen through the flaring portion of the seating portion **810**).

In some embodiments, placing the module rail in the seating portion **810** of the module clamp **800** may apply an outward force to the peg **832** of the spring-loaded protrusion **830**, causing the peg **832** to retract into a spring chamber **834** of the spring-loaded protrusion **830**. When a hole in the module rail is aligned with the peg **832**, the spring force of the spring chamber **834** may cause the peg **832** to spring back out of the spring chamber **834** and into the hole in the module rail. The peg **832** being disposed in the hole of the module rail and biased into the hole of the module rail by the spring force of the spring chamber **834**, may lock the module rail in position relative to the module clamp **800**.

In some embodiments, the module clamp **800** may include a fastener **840** extending through a base surface of the seating portion **810**. In some embodiments, the fastener **840** may be configured to interface with a corresponding opening in the module rail (e.g., on a bottom surface of the module rail configured to sit flush against the base surface of the seating portion). Additionally or alternatively, the fastener **840** may be configured to interface with a surface of the mounting rail to reduce and/or prevent sliding, rattling, shifting, or any other movements of the mounting rail.

Additionally, FIG. **8** illustrates examples of a nut and bolt arrangement **850** of the protruding feature and a rivet example **860** of the protruding feature. As illustrated in FIG. **8**, in some embodiments multiple types of protruding features may be disposed in a single implementation. For example, spring-loaded protrusions **830** may be used on one side of the seating portion **810** while rivets or bolted pegs may be disposed on the other side of the seating portion **810**.

FIG. **9** illustrates a side view of a custom bolt **900** according to at least one embodiment of the present disclosure. In some embodiments, the custom bolt **900** may be used as the protruding feature (e.g., in place of the protruding nubs **330**, the insertion pegs **430**, the external wire forms **530**, the spring tabs **630**, the hook tabs **730**, and/or the spring-loaded protrusion **830** as described above). The custom bolt **900** may include a head **910**, a body **920**, and a custom nut **930**. The body **920** may be inserted through a surface **950**, such as the flared portion of the lateral walls of the seating portion, causing the head **910** to be positioned flush against a first side of the surface **950**. The custom nut **930** may be coupled to the body **920** such that the custom nut **930** is positioned flush against a second side of the surface **950**. In some embodiments, the custom nut **930** may include one or more sloped edges **940** (e.g., chamfer edges) and/or protrusions with sloped edges, which may facilitate spreading out the lateral walls of the module clamp during installation of the module rail with the module clamp. Additionally or alternatively, the custom nut **930** may facilitate preventing the module rail from dislodging from the custom nut **930** acting as the protruding feature. For example, the protrusions at the distal end of the custom nut **930** may catch or otherwise prevent the module rail from moving past the protrusions.

FIG. **10**A is an isometric view of a module clamp **1000** according to at least one embodiment of the present disclosure, and FIG. **10**B is a side view of the module clamp **1000**. The module clamp **1000** may include one or more spacers **1010** positioned against a clamp body **1005**. For example, the spacers **1010** may be positioned flush against a bottom-left surface and/or a bottom-right surface of the clamp body **1005** as illustrated in FIGS. **10**A and **10**B. Additionally or alternatively, the spacers **1010** may be positioned flush against a left surface, a right surface, a bottom surface, a top surface, a top left surface, a top-right surface, or any other surface of the clamp body **1005**. Additionally or alternatively, the spacers **1010** may include a shape and/or size corresponding to the geometry of the clamp body **1005** and/or the module support structure (e.g., torque tube). For example, the spacers **1010** may include a curved shape to correspond to a curved surface of the clamp body **1005** and/or the module support structure (e.g., torque tube). As illustrated in FIG. **10**B, the spacers **1010** coupled to the clamp body **1005** may be positioned flush against one or more surfaces of a module support structure. In some embodiments, the spacers **1010** may reduce and/or prevent movement of the module support structure and/or a corresponding PV module coupled to the module clamp **1000**. Additionally or alternatively, the spacers **1010** may facilitate coupling the module clamp **1000** to which the spacers **1010** are attached to a wider variety of module support structures including different sizes and/or geometries. In these and other embodiments, the spacers **1010** may include a textured or roughed surface to increase friction between the surfaces of the spacers **1010** and the surfaces of the module support structure. Additionally or alternatively, the spacers **1010** may facilitate ensuring surface-to-surface contact between the module support structure (e.g., torque tube) and the PV module rail.

FIG. **11** illustrates a seating portion **1105** of a module clamp **1100** according to at least one embodiment of the present disclosure. The seating portion **1105** may include various opening patterns, such as patterns **1110**-**1140**, to facilitate fitting various shapes, sizes, and/or configurations of module rails. In some embodiments, one or more lateral walls **1150** of the seating portion **1105** may include a sloped profile (not shown) such that placing the mounting rail in the seating portion **1105** causes one or more surfaces of the mounting rail to follow the sloped profile. Additionally or alternatively, the lateral walls **1150** having the sloped profiles may include one or more increases in height (e.g., curved hills, straight-edged plateaus, or any other sloped profiles) towards the interior region of the seating portion **1105** followed by a subsequent decrease in height away from the interior region of the seating portion **1105**, which may increase the force used to insert the mounting rail in the seating portion **1105** and further increase the force used to remove the mounting rail from the seating portion **1105** in the same direction.

FIGS. **12**A and **12**B are isometric views of an eighth example embodiment of a module clamp **1200** that may be coupled to a torque tube **1210** according to at least one embodiment of the present disclosure. The module clamp **1200** may include a seating portion **1220** that has one or more pop-out tabs **1230** that may be cut out, punched out, or otherwise formed from the lateral walls of the seating portion of the snap-lock PV module mounting system. In some embodiments, the pop-out tabs **1230** may be fastening features oriented such that the cut-out sections face the open end of the seating portion (as illustrated in FIGS. **12**A and **12**B), projecting into the interior region of the seating portion **1220**. In these and other embodiments, placing the module rail in the seating portion **1220** may push the lateral walls of the module clamp outward and facilitate the pop-out tabs **1230** interfacing with one or more corresponding openings, hooks, and/or any other structures of the module rail. Additionally or alternatively, the module rail and/or the seating portion **1220** may be sized to be comparable in size such that the lateral walls of the seating portion **1220** may not be displaced by the module rail being pressed into the seating portion **1220** but may push against the pop-out tabs **1230** until one or more mounting structures along the module rail are aligned with openings **1232** on the pop-out tabs **1230** and/or the pop-out tabs **1230**. When aligned, the mounting structures of the module rail may extend through the openings **1232** and cause the pop-out tabs **1230** to fasten module rail to the module clamp **1200**. Additionally or alternatively, the pop-out tabs **1230** may be the portion of the seating portion **1220** that is forced to flex out of the way when the module rail is pressed into the seating portion, and the flexion of the pop-out tabs **1230** may cause the pop-out tabs **1230** to pop back into an opening in the module rail when aligned with the pop-out tabs.

FIG. **13** is a close-up view of a module clamp **1300** of the snap-lock PV module mounting system according to at least one embodiment of the present disclosure. The module clamp **1300** may include a seating portion **1310** that is shaped such that a module rail of a PV module may be aligned to sit within the seating portion **1310**. The seating portion **1310** may be made of one or more lateral walls **1312** that are connected by a base surface **1314**. In some embodiments, the lateral walls **1312** and the base surface **1314** may form a U-shaped channel into which the module rail of the PV module may be positioned. In these and other embodiments, the seating portion **1310** of the module clamp **1300** may be the same as or similar to the seating portion **144** described in relation to FIG. **1**B.

In some embodiments, one or more tabs **1330** may be fastening features included on the lateral walls **1312** of the seating portion **1310**. The tabs **1330** may be positioned along a length of the lateral walls **1312** and protrude into an interior of the seating portion **1310** such that the tabs **1330** align with holes, hooks, protrusions, or any other mounting features of a module rail that is placed in the seating portion **1310**. As the module rail is placed in the seating portion **1310**, the lateral walls **1312** may flare out and expand the space of the seating portion **1310** to accommodate the module rail, and after being placed a sufficient distance within the seating portion **1310**, the tabs **1330** may interface with the mounting features (e.g., the openings) of the module rail to secure the module rail relative to the module clamp **1300**.

In some embodiments, the module clamp **1300** may include a pressure pad **1320** that is made of one or more bridge lances **1322**, **1324**, which may protrude from the base surface **1314** of the seating portion **1310**. In some embodiments, an example module rail, when fully seated within the seating portion **1310**, may not completely reach the base surface such that a bottom surface of the example module rail interfaces with the base surface **1314**. The pressure pad **1320** may provide a deformable surface against which the module rail may be pressed when seating into the seating portion **1310** that provides a spring force on the module rail positioned on the base surface **1314**. In these and other embodiments, the pressure pad **1320** may be positioned underneath the tabs **1330** such that pressing the module rail into the seating portion **1310** deforms the bridge lances **1322**, **1324** and after some deformation, the openings of the module rail interface with the tabs **1330** decreases deformation of the bridge lances **1322**, **1324**. Stated another way, after the openings of the module rail interface with the tabs **1330**, the spring force of the bridge lances **1322**, **1324** may force the module rail upwards against the tabs **1330**. Such a feature may lock the module rail in position more securely and may prevent rattling or movement of the module rail within the seating portion **1310**. Using such a feature may utilize an increased force when pressing the module rail into the seating portion **1310** when compared to a seating portion **1310** that may not include the pressure pad **1320**.

FIG. **14**A is a close-up view of a module clamp **1400** of a snap-lock PV module mounting system according to at least one embodiment of the present disclosure. In some embodiments, the module clamp **1400** may be the same as or similar to the module clamp **1300** as described in relation to FIG. **13** in that the module clamp **1400** includes lateral walls **1412** and a base surface **1414** that form a seating portion **1410***a* in which a mounting rail may be placed. To secure the mounting rail relative to the seating portion **1410***a*, the module clamp **1400** may include one or more openings through which a fastener **1430** may be slotted. In some embodiments, the fastener **1430** may be slotted through the openings of the module clamp **1400** to lock the module rail relative to the module clamp **1400**.

Additionally or alternatively, the module clamp **1400** may include an extended tab **1420** that protrudes from the base surface **1414**, and the module rail placed in the seating portion **1410***a* may rest on the extended tab **1420**. In some embodiments, the extended tab **1420** may include a trapezoidal shape or any other hill shape so that a top surface **1422** of the extended tab **1420** is higher than the base surface **1414** of the seating portion **1410***a*. The extended tab **1420** may be depressed by the module rail, and the shape of the extended tab **1420** may provide a spring force that opposes the downward force of the module rail being placed in the seating portion **1410***a*.

In some embodiments, when the module rail is resting on the top surface **1422**, the openings in the module rail may not align with the holes through which the fastener **1430** is configured to extend. When the module rail is pressed against the top surface **1422**, causing the extended tab **1420** to deform, the openings in the module rail and the lateral walls **1412** may align such that the fastener **1430** may then be fit into the openings. This may cause the extended tab **1420** to be in a deformed position when the fastener is in place, which may provide a spring force to keep the module rail and the seating portion **1410***a* locked into position such that the two move as a single component.

FIG. **14**B is a close-up view of a module clamp **1450** of a snap-lock PV module mounting system according to at least one embodiment of the present disclosure. In some embodiments, the module clamp **1450** may be similar to the module clamp **1400** in that the module clamp **1450** includes the lateral walls **1412** and the base surface **1414** that form a seating portion **1410***b* in which a PV module rail **1460** may be placed. The module clamp **1450** may include a sump protrusion **1470** along the base surface **1414** of the seating portion **1410***b*. In some embodiments, the sump protrusion **1470** may be configured such that the PV module rail **1460** rests flush against the sump protrusion **1470** when the PV module rail **1460** is positioned in the seating portion **1410***b*. Consequently, the PV module rail **1460** may be offset from the base surface **1414** by a height of the sump protrusion **1470** when resting in the seating portion **1410***b* of the module clamp **1450**. In these and other embodiments, the sump protrusion **1470** may provide an opposing force against the PV module rail **1460** that is not a spring force as provided by the bridge lances **1322**, **1324** of the module clamp **1300** illustrated in FIG. **13** and/or the extended tab **1420** of the module clamp **1400** illustrated in FIG. **14**A. In these and other embodiments, the sump protrusion **1470** may be sized to provide spacing such that openings in the PV module rail **1460** may align with openings in the lateral walls **1412** of the seating portion **1410***b*.

FIG. **15** is an isometric view of a module clamp **1500** according to at least one embodiment of the present disclosure. The module clamp **1500** may include a clamp body **1515** and/or a seating portion **1510** that are the same as or similar to the clamp bodies and/or seating portions of other embodiments of the snap-lock PV module mounting systems described in relation to, for example, FIGS. **3**A-**3**C, **4**A-**4**C, **5**A-**5**E, **6**, and/or **12**A-**12**B. For example, the seating portion **1510** of the module clamp **1500** may include lateral walls **1512** and a base surface **1514**.

The seating portion may include one or more pop-out tabs **1530** that may be fastening features similar or comparable in function to the pop-out tabs of FIGS. **12**A-**12**B. However, the pop-out tabs **1530** may be made of a different material or as a separate component from the lateral walls **1512**. For example, the pop-out tabs **1530** may be made of a rigid plastic or a spring metal and may be riveted, welded, melted, adhered, bolted, or otherwise fixedly attached to the lateral walls **1512**.

In some embodiments, the module clamp **1500** may include one or more pressure pads **1520** that protrude from the base surface **1514**, and a module rail positioned within the seating portion **1510** may be seated on the pressure pads **1520** with a gap between the module rail and the base surface **1514**. The pressure pads **1520** may provide a spring force that opposes the module rail seated on the pressure pads **1520** within the seating portion **1510**.

In these and other embodiments, each of the pressure pads **1520** may include a pad component **1522** connected to a screw or other body component **1524** that is coupled to a cam lever **1526**, and a height of the pressure pads **1520** may be adjusted by a cam lever **1526** coupled to the pressure pads **1520**. Rotating the cam lever **1526** may adjust a length of the body component **1524** above the base surface **1514**, which corresponds to a height of the pressure pad **1520** above the base surface **1514**. Additionally or alternatively, one end of the body component **1524** may include an adjustable knob, such as a screwhead, that may be adjusted to change the length of the body component **1524** above the base surface **1514** in tandem with or independent of rotation of the cam lever **1526**. Although illustrated and described as a pad component **1522**, the pressure pads **1520** may include a set screw, a knob, a spring, or any other component in addition to or as a substitute for the pad component **1522** to provide the opposing force against the module rail positioned on top of the pressure pads **1520**.

As with the extended tab **1420** of FIG. **14**A, the pressure pad **1520** of FIG. **15** may press the module rail against the pop-out tabs **1530** as the fastening feature. For example, the pressure pad may have the cam lever **1526** released when pressing the module rail into the seating portion **1510** such that openings of the module rail align with the pop-out tabs **1530**. After being positioned, the cam levers **1526** may be engaged, forcing the module rail upwards and against the pop-out tabs **1530**, locking the module rail and the seating portion **1510** in a more secure connection relative to each other to avoid or prevent sliding, rattling, or movement between the two components.

FIGS. **16A-16C** illustrate various views of a module clamp **1600** that couples to a torque tube **1630** via one or more wire module clamps **1610** according to at least one embodiment of the present disclosure. The module clamp **1600** may include one or more wire module clamps **1610** that fasten the torque tube **1630** to a module rail **1620**. In some embodiments, the wire module clamps **1610** may be fastening features made of one or more wire frames **1612** that connect hooks **1618** to a fastening loop **1614**. The wire frames **1612**, the hooks **1618**, and/or the fastening loop **1614** may include angles or a series of portions that are shaped to interface with a profile of the torque tube **1630**.

In these and other embodiments, two or more wire module clamps **1610** may be positioned on opposite sides of the torque tube **1630** so that the hooks **1618** may interface with one or more openings and/or holes along opposite sides of the module railing, and a bolt, screw, or any other fastener may be inserted through the fastening loops **1614** of each of the wire module clamps **1610** to force the wire module clamps **1610** against the torque tube **1630** and the torque tube **1630** against the module rail. For example, a pair of such wire frames **1612** may be used with one on either side of the torque tube **1630** and a bolt **1616** may connect the two along the bottom of the torque tube **1630**. In this and other examples, the bolt **1616** may be tightened to pinch the wire frames **1612** together to provide a stronger locking force between the module rail, module clamp **1600**, and/or the torque tube **1630**.

The wire frames **1612** may be attached to the module rail **1620** prior to installation and prior to shipping such that when an installer arrives to install a PV tracking system, the PV module rails may already include the wire frames **1612** attached thereto without taking much, if any, additional space in shipping or packaging. Additionally or alternatively, the wire frames **1612** may be attached to the module rail on-site during installation.

In installation, the wire frames **1612** may be attached to the module rail **1620** (whether in the factory during manufacturing or in the field) and the module rail **1620** and corresponding PV modules may be placed on the torque tube. The wire frames **1612** may be placed in proximity to each other (effectively pinching the torque tube between the two wire frame **1612** components), and the bolt **1616** may be tightened between the two wire frames **1612**, locking the module rail **1620** in position relative to the torque tube.

In some embodiments, the height/profile of the wire frames **1612** may be the same or smaller in profile than the module rail **1620**. In these and other embodiments, the wire frames **1612** may be attached to the module rail **1620** and may rotate back in place or forward in place against the back of the PV module(s) such that the wire frames **1612** may be lower than the module rail **1620**. In these and other embodiments, stacking and/or shipping module rail **1620** and PV modules may or may not require additional space when shipping the module rail **1620**/PV modules to a site for installation with the wire frames **1612** already installed on the module rail **1620**.

Terms used in the present disclosure and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open terms" (e.g., the term "including" should be interpreted as "including, but not limited to.").

Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (e.g., "a" and/or "an" should be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is expressly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of "two recitations," without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, etc." or "one or more of A, B, and C, etc." is used, in general such a construction is intended to include A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together, etc.

Further, any disjunctive word or phrase preceding two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both of the terms. For example, the phrase "A or B" should be understood to include the possibilities of "A" or "B" or "A and B."

All examples and conditional language recited in the present disclosure are intended for pedagogical objects to aid the reader in understanding the present disclosure and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present disclosure have been described in detail, various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the present disclosure.

What is claimed:

1. A mounting system for photovoltaic (PV) modules comprising:

a module rail configured to be secured to a PV module, the module rail having a first lateral wall, a second lateral wall, and a base portion that connects at least portions of bottom edges of the first and second lateral walls;

a first wire form module clamp having a first end, a second end, and a curve forming a first fastening loop between the first and second ends, wherein the first and second ends are secured to the first and second lateral walls of the module rail;

a second wire form module clamp having a first end, a second end, and a curve forming a second fastening loop between the first and second ends, wherein the first and second ends are secured to the first and second lateral walls of the module rail; and

a fastener that couples the first fastening loop to the second fastening loop, such that in a coupled configuration, the first and second wire form module clamps are configured to secure the module rail to a torque tube;

wherein:

to secure the first wire form module clamp to the module rail, the first and second ends of the first wire form module clamp are inserted through holes defined by the first and second lateral walls, respectively; and

to secure the second wire form module clamp to the module rail, the first and second ends of the second wire form module clamp are inserted through holes defined by the first and second lateral walls, respectively.

2. The mounting system of claim **1**, wherein the first and second wire form module clamps are pivotally secured to the module rail such that a distance between the first and second fastening loops may be selectively adjusted.

3. The mounting system of claim **1**, wherein hook shapes are formed in the first and second ends of the first and second wire form module clamps to maintain the first and second wire form module clamps within the holes defined by the first and second lateral walls.

4. The mounting system of claim **1**, wherein the first and second wire form module clamps each comprise a single length of wire having a circular cross-sectional shape.

5. The mounting system of claim **1**, wherein the first and second wire form module clamps are shaped to interface with an outer surface of the torque tube.

6. The mounting system of claim **5**, wherein the first and second wire form module clamps include bends that are consistent with a polygonal shaped torque tube.

7. The mounting system of claim **1**, wherein the fastener can be selectively tightened to decrease a distance between the first and second fastening loops and increase a locking force between the first and second wire form module clamps and the torque tube.

8. The mounting system of claim **7**, wherein the fastener includes a bolt.

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